EDUVERSE AI
SMART COLLEGE RECOMMENDATION SYSTEM FOR TAMIL NADU

PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree of
MASTER OF COMPUTER APPLICATIONS

[COLLEGE EMBLEM / LOGO PLACEHOLDER]

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KCG COLLEGE OF TECHNOLOGY
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Certified that this Project Report titled "EDUVERSE AI – SMART COLLEGE RECOMMENDATION SYSTEM FOR TAMIL NADU" is the bonafide work of PRIYADHARSHINI R (Roll No: 9525621125) and PRIYANKA T (Roll No: 9525621126) who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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We further declare that this report has not previously formed the basis for the award of any degree, diploma, associateship, fellowship, or any other similar title to any candidate of any university or institution.

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ABSTRACT

Transitioning from secondary education to higher education represents one of the most critical and complex decisions in a student's academic journey. In Tamil Nadu, the higher education landscape encompasses over a thousand engineering, arts, and science institutions distributed across 38 districts, comprising government, government-aided, and private self-financing establishments. Students and parents frequently encounter severe information asymmetry, commercial ranking bias on aggregator portals, fragmented data sources, and confusion regarding TNEA engineering cutoff calculations. Commercial portals routinely prioritize sponsored listings over objective institutional quality, while public counseling websites present dense, unstructured tabular data that lacks personalized guidance and intuitive decision support.

To resolve these structural bottlenecks, this project presents EduVerse AI – a full-stack, AI-augmented college discovery, comparison, and recommendation web platform designed specifically for higher education aspirants in Tamil Nadu. Developed using Python Flask 3.1.0, SQLAlchemy 2.0 ORM, SQLite 3/MySQL relational database architectures, and responsive HTML5/CSS3 frontend components, EduVerse AI indexes 1,129 verified colleges and 8,120 courses covering all 38 districts of Tamil Nadu. The platform incorporates multi-parametric faceted search, side-by-side institutional comparison, direct verified official website routing, and an interactive 12th standard academic assessment tool that computes official Tamil Nadu TNEA engineering cutoffs on both 100-mark and standardized 200-mark counseling scales.

The core intelligent advisory engine integrates an explainable Content-Based Recommendation System with multi-attribute scoring evaluating target degree, district proximity, tuition fee ceilings, institutional type, and campus amenities. Furthermore, a resilient AI Counselor service combines Google Gemini generative capabilities with a zero-downtime deterministic rule-based counseling engine. Empirical validation through 85 automated pytest test cases confirms 100% test pass rates and sub-50ms endpoint latencies, delivering a reliable, transparent, and scalable educational decision-support ecosystem.

சுருக்கம் (TAMIL ABSTRACT)

பன்னிரண்டாம் வகுப்பு பொதுத்தேர்வை முடித்த மாணவர்கள் தங்களின் அடுத்தகட்ட உயர்கல்வி மற்றும் கல்லூரிப் பிரிவை தேர்ந்தெடுப்பது என்பது அவர்களின் எதிர்கால வாழ்க்கையை தீர்மானிக்கும் மிக முக்கியமான முடிவாகும். தமிழ்நாட்டில் உள்ள 38 மாவட்டங்களில் ஆயிரத்திற்கும் மேற்பட்ட அரசு, அரசு உதவிபெறும், தனியார் சுயநிதி மற்றும் தன்னாட்சிப் (Autonomous) பொறியியல், கலை மற்றும் அறிவியல் கல்லூரிகள் இயங்கி வருகின்றன. இத்தகைய பரந்த கல்விச் சூழலில், மாணவர்கள் தங்களின் மதிப்பெண்கள், ஆர்வங்கள் மற்றும் இருப்பிடத்திற்கு ஏற்ற சிறந்த கல்லூரிகளை கண்டறிவதில் போதிய விழிப்புணர்வின்மையும், சரியான வழிகாட்டலின்மையும் பெரும் சவாலாக உள்ளது. பல வணிக நோக்குடைய இணையதளங்கள் முழுமையற்ற மற்றும் விளம்பரச் சார்புடைய தகவல்களையே வழங்குகின்றன.

இச்சிக்கல்களுக்குத் தீர்வாக, EduVerse AI என்ற முழுமையான, செயற்கை நுண்ணறிவு (Artificial Intelligence) சார்ந்த உயர்கல்வி வழிகாட்டுதல் மற்றும் கல்லூரி ஒப்பீட்டு வலைப்பின்னல் தளம் உருவாக்கப்பட்டுள்ளது. இத்தளம் பைதான் பிளாஸ்க் (Python Flask) கட்டமைப்பிலும், SQL தரவுத்தள (Database) அடிப்படையிலும், HTML, CSS மற்றும் ஜாவாஸ்கிரிப்ட் (JavaScript) தொழில் நுட்பங்களுடனும் கட்டமைக்கப்பட்டுள்ளது. இதில் தமிழ்நாட்டின் அனைத்து 38 மாவட்டங்களிலும் உள்ள 1,129 அங்கீகரிக்கப்பட்ட கல்லூரிகள் மற்றும் 8,120 பாடப்பிரிவுகளின் அதிகாரப்பூர்வ விபரங்கள், அமைவிடம் மற்றும் சரிபார்க்கப்பட்ட அதிகாரப்பூர்வ இணையதள முகவரிகள் (Official Websites) ஒருங்கிணைக்கப்பட்டுள்ளன.

மாணவர்கள் தங்கள் மாவட்டங்கள், கட்டண வரம்புகள், தன்னாட்சி அந்தஸ்து மற்றும் பாடப்பிரிவுகளின் அடிப்படையில் கல்லூரிகளை தேடவும் (Search & Filter), பல கல்லூரிகளை ஒரே திரையில் ஒப்பிட்டுப் பார்க்கவும் (College Comparison), மாவட்ட வாரியான கல்வி புள்ளிவிபரங்களை (Analytics) காணவும் முடியும். மேலும், 12-ஆம் வகுப்பு மதிப்பெண்களை உள்ளீடு செய்து TNEA பொறியியல் கலந்தாய்வு Cutoff மதிப்பெண்களை கணக்கிடும் முறையும், விருப்பங்களின் அடிப்படையில் கல்லூரிகளை வரிசைப்படுத்தி பரிந்துரைக்கும் (Recommendation Algorithm) அமைப்பும் இதில் நிறுவப்பட்டுள்ளது. கூகிள் ஜெமினி (Google Gemini API) மற்றும் விதிமுறை சார்ந்த ஆலோசனை இயந்திரம் (Rule-Based Counselor Engine) மூலம் மாணவர்கள் தங்களுக்குப் பொருத்தமான உயர்கல்வி மற்றும் வேலைவாய்ப்பு ஆலோசனைகளை நேரடியாகப் பெற முடிகிறது. இத்திட்டத்தின் செயல்முறைகள் 85 தானியங்கி சோதனைகள் (Pytest) மூலம் வெற்றிகரமாக சரிபார்க்கப்பட்டு 100% துல்லியத்துடன் இயங்குவது உறுதி செய்யப்பட்டுள்ளது.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AISHE	All India Survey on Higher Education
API	Application Programming Interface
CBF	Content-Based Filtering
CSRF	Cross-Site Request Forgery
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DOM	Document Object Model
DTE	Directorate of Technical Education, Tamil Nadu
ER	Entity Relationship
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MCA	Master of Computer Applications
NAAC	National Assessment and Accreditation Council
NIRF	National Institutional Ranking Framework
ORM	Object-Relational Mapping
REST	Representational State Transfer
SaaS	Software as a Service
SQL	Structured Query Language
TF-IDF	Term Frequency - Inverse Document Frequency
TNEA	Tamil Nadu Engineering Admissions
TNDCE	Tamil Nadu Directorate of Collegiate Education
UGC	University Grants Commission
UI/UX	User Interface / User Experience
UML	Unified Modeling Language
URI/URL	Uniform Resource Identifier / Locator
WSGI	Web Server Gateway Interface
XSS	Cross-Site Scripting

CHAPTER 1 – INTRODUCTION

1.1 Background of the Study

The transition from higher secondary schooling (12th standard) to tertiary higher education constitutes a pivotal milestone in a student's life. The choices made at this juncture significantly influence long-term career trajectories, intellectual growth, and socio-economic mobility. Within the Indian educational framework, and specifically within the state of Tamil Nadu, higher education is characterized by extensive institutional diversity and institutional volume. The state hosts prominent central institutes, state universities, government constituent colleges, government-aided institutions, private self-financing colleges, and deemed-to-be universities. These institutions are distributed across 38 administrative districts, encompassing thousands of approved undergraduate programs in engineering, technology, computer applications, arts, science, commerce, and vocational studies.

In recent years, the democratization of internet access has created an expectation that information regarding colleges, courses, facilities, and cutoff eligibility should be readily accessible to every student, irrespective of geographic or economic background. However, the ground reality often presents an acute paradox: while information is abundant across the internet, reliable, consolidated, and objective institutional intelligence remains remarkably difficult to obtain.

Students concluding their 12th standard board examinations face several critical hurdles:

1. **Dispersed and Inconsistent Data:** College information is scattered across hundreds of separate portals, varying state government circulars (such as the Directorate of Technical Education and Directorate of Collegiate Education), autonomous university portals, and unregulated commercial listing sites.

2. **Commercial Bias and Sponsored Content:** Commercial aggregator portals commonly prioritize sponsored institutions, displaying promoted colleges rather than matching student merit and preferences.

3. **Ambiguity in Admission Formulas and Cutoffs:** Many students struggle to calculate their normalized Tamil Nadu Engineering Admissions (TNEA) cutoff or understand how their 12th standard subject aggregate aligns with collegiate eligibility criteria.

4. **Lack of Verified Institutional Web Portals:** Phishing, duplicate, and stale web links proliferate online, exposing students to misdirection when attempting to visit an institution's genuine administrative or admission portal.

5. **Absence of Explainable Guidance:** Traditional portals present rigid database listings without personalized, multi-attribute recommendation algorithms that explain *why* a particular institution or course is suitable for a given student.

Modern web technologies and Artificial Intelligence provide effective avenues to overcome these structural deficiencies. By implementing a modern full-stack web architecture with a lightweight backend, an expressive Object-Relational Mapper, a structured database, and conversational AI services, it is possible to deliver an objective, transparent, and student-centered guidance ecosystem.

1.2 Problem Statement

Despite the proliferation of digital directories, higher secondary school graduates in Tamil Nadu lack a unified, trustworthy, and intelligent web platform to systematically discover, compare, and evaluate higher education opportunities.

Specifically, the core problems addressed by this project include:

- **Absence of a Consolidated Tamil Nadu Higher Education Directory:** No unified, non-commercial platform systematically catalogues colleges across all 38 districts of Tamil Nadu with multi-dimensional taxonomy (Government, Government-Aided, Self-Financing, Autonomous status) and verified official website endpoints.

- **Difficulty in Side-by-Side Institutional Comparison:** Students lack intuitive tools to juxtapose multiple colleges simultaneously across standardized parameters such as annual fee ceilings, academic affiliations, course offerings, accreditation ratings, and infrastructure facilities.

- **Confusion in Cutoff Computation and Eligibility Assessment:** Students from diverse academic streams (Science, Computer Science, Commerce, Arts) frequently struggle to calculate their aggregate percentages and official TNEA cutoff formulas, leading to unrealistic or misguided application submissions.

- **Opacity in College Recommendations:** Existing filtering tools operate as binary search filters without intelligent multi-criteria scoring models that evaluate course relevance, geographic location affinity, budget compatibility, and campus facilities while providing explainable match justifications.

- **Limited Access to Personalized Career and Academic Counseling:** Personalized educational guidance from qualified human counselors is cost-prohibitive for many rural and economically disadvantaged students, creating a pressing need for automated, interactive AI counseling grounded in Tamil Nadu higher education pathways.

1.3 Objectives of the Project

The primary objective of **EduVerse AI** is to design, implement, and validate a full-stack, AI-powered college and course guidance web platform that empowers students completing 12th standard in Tamil Nadu to make well-informed, data-driven academic decisions.

The specific, measurable technical objectives are:

1. **Architect a Scalable Full-Stack Web Application:** Implement a modular Flask application following the Application Factory design pattern, organized into cohesive Blueprints representing discrete functional domains.

2. **Build and Curate a Verified Collegiate Database:** Establish a structured relational database indexing 1,129 verified colleges and 6,290 courses spanning all 38 districts of Tamil Nadu, complete with geocoordinates, official institutional website URLs, and accreditation data.

3. **Develop a Multi-Attribute Search and Filtering Engine:** Deliver responsive search and filtering capabilities supporting queries across district, college type, autonomous status, course specialization, and tuition fee boundaries.

4. **Engineer an Interactive College Comparison Matrix:** Implement a session-backed comparison module allowing students to compare candidate institutions side-by-side across courses, fees, accreditations, and campus amenities.

5. **Implement an Academic Assessment and TNEA Cutoff Engine:** Formulate automated calculation services that parse 12th standard marks across academic streams, calculate aggregate percentages, and accurately compute standardized TNEA engineering cutoffs.

6. **Deploy an Explainable Content-Based Recommendation System:** Formulate a multi-criteria scoring algorithm (evaluating course match 35%, location affinity 25%, budget ceiling 20%, college type 10%, and campus facilities 10%) that outputs ranked recommendations with transparent, human-readable match factors.

7. **Integrate an AI Education Counselor with Graceful Fallback:** Embed an interactive counseling service connecting to the Google Gemini API with student profile contextualization, supported by an autonomous rule-based guidance engine to guarantee continuous availability during network disruptions.

8. **Provide Dynamic Educational Analytics:** Develop an interactive analytics dashboard displaying real-time distributions of institutions, course categories, and district representations using Chart.js.

9. **Ensure Production-Grade Code Quality and Security:** Verify the system through automated unit and integration tests (pytest), CSRF protection, secure URL sanitization, and SQL injection prevention via parameterized ORM queries.

1.4 Scope of the Project

The operational boundaries and functional inclusions of EduVerse AI are defined as follows:

Included within Scope:

- Coverage of all 38 administrative districts of Tamil Nadu.

- Indexing of higher education institutions spanning Government, Government-Aided, Private Self-Financing, Autonomous, and University Departments.

- Course specializations covering Undergraduate Engineering (B.E./B.Tech), Computer Applications (BCA), Computer Science & Information Technology (B.Sc CS/IT), Commerce (B.Com), Business Administration (BBA), and Arts/Humanities.

- Standardized TNEA engineering cutoff computation using the official Tamil Nadu formula.

- Verification and safe routing to 100% genuine institutional official websites with URL sanitization.

- Role-based access control for Students and Administrators.

- Administrative management portal supporting CRUD operations on colleges, courses, facilities, student queries, and system audit logs.

- Interactive AI counselor with Google Gemini API integration and rule-based fallback logic.

- Automated testing suite verifying routes, services, and recommendation scoring.

Explicitly Outside Current Scope (Operational Boundaries):

- Real-time seat matrix availability during live Anna University counseling rounds (requires direct TNEA server integration).

- Direct processing of collegiate admission application fee transactions.

- Post-graduate entrance examination counseling (e.g., TANCET, GATE, CAT).

- Hardware-level biometric or SMS OTP authentication gateways (uses email/password session authentication).

- Native iOS/Android mobile binaries (system delivers full responsive mobile web capabilities).

1.5 Organisation of the Report

The subsequent chapters of this academic project report are structured as follows:

- **Chapter 2 – Literature Review:** Reviews historical paradigms in online college discovery, recommender systems in education, artificial intelligence applications in academic advising, and formulates research gaps.

- **Chapter 3 – System Analysis:** Examines the limitations of existing approaches, defines the proposed system, documents functional and non-functional requirements, specifies hardware and software prerequisites, and presents feasibility and constraint analyses.

- **Chapter 4 – System Design:** Details the overall architecture, layered design, UML use case modeling, Data Flow Diagrams (Context & Level-1), application workflow, relational database schema, RESTful API catalog, and user interface design.

- **Chapter 5 – System Implementation:** Explains the development environment, project directory organization, Flask initialization, data access pipelines, search and comparison algorithms, recommendation scoring mathematics, AI counseling service, security implementations, and core code excerpts.

- **Chapter 6 – Testing and Performance Evaluation:** Reports the testing methodology, automated test suite execution (pytest), live API latency benchmarks, browser compatibility, security auditing, and presents application interface screenshots.

- **Chapter 7 – Conclusion and Future Enhancements:** Concludes the project outcomes, delineates verified operational limitations, and identifies systematic future enhancements.

- **Appendices & References:** Provides core source code excerpts, API catalogs, database schemas, installation manuals, dependency manifests, and academic references.

CHAPTER 2 – LITERATURE REVIEW

2.1 Evolution of Online College Discovery Systems

The transition from print-based institutional prospectuses to web-based educational directories represents a significant evolution in educational informatics. In early digital paradigms, college discovery relied on static institutional homepages and rudimentary directory listings published by governmental bodies such as the All India Survey on Higher Education (AISHE) and the National Assessment and Accreditation Council (NAAC). While authoritative, these early repositories functioned primarily as administrative registries rather than interactive decision-support platforms (Kumar & Sharma, 2018).

With the growth of Web 2.0, commercial education aggregators emerged to address the user experience limitations of governmental databases. Portals such as Shiksha, Careers360, and CollegeDunia introduced multi-attribute search filters, user-submitted campus reviews, and centralized brochure downloads. However, contemporary scholarship has highlighted notable structural deficiencies in commercial aggregators. Patel and Roy (2021) demonstrated that commercial platforms frequently introduce algorithmic bias by prioritizing educational institutions that purchase sponsored placement packages. This prioritization displaces merit-based or cost-optimal options from prominent search tiers. Furthermore, Aggarwal et al. (2020) observed that third-party portals frequently fail to maintain synchronized hyperlinks to genuine institutional portals, leading users to broken URLs or redirecting student inquiries to external lead-generation telemarketers. EduVerse AI addresses this historical limitation by maintaining an independent, non-commercial directory with direct, verified hyperlinks to official collegiate domains.

2.2 Education Recommendation Systems

Recommender systems have been widely investigated across commercial domains including e-commerce, entertainment streaming, and social networks. Their application within educational decision support, however, necessitates significantly higher precision and transparency due to the life-altering consequences of academic choices (Manouselis et al., 2011).

In educational recommendation literature, three primary algorithmic paradigms predominate:

1. **Collaborative Filtering (CF):** Predicts student preference based on the historical choices of peer cohorts with similar academic profiles (Su & Khoshgoftaar, 2009). While effective in consumer retail, CF suffers from severe "cold-start" deficiencies in collegiate selection. Graduating 12th standard students enter the system as first-time users without historical transaction records, making pure collaborative filtering ineffective for initial guidance (Burke, 2002).

2. **Knowledge-Based and Multi-Criteria Recommendation:** Leverages explicit domain knowledge, academic constraint satisfaction, and multi-attribute utility theory (MAUT) to evaluate items against user-defined parameters (Felfernig et al., 2014). This approach is highly resilient to cold-start phenomena because it relies on explicit student inputs (e.g., marks, budget, location preferences) rather than historical community ratings.

3. **Content-Based Filtering:** Recommends items by matching user profile vectors with feature representations of candidate items (Lops et al., 2011). In collegiate selection, institutional feature vectors comprise course offerings, geographic coordinates, tuition fee distributions, and accreditation credentials.

EduVerse AI adopts a hybrid knowledge-based and content-based recommendation approach. By evaluating candidate institutions across a weighted multi-attribute scoring framework, the system generates ranked recommendations that are transparent and explainable.

2.3 Artificial Intelligence in Student Decision Support

The integration of Artificial Intelligence into academic advising has transitioned from rigid expert systems of the 1980s to modern Large Language Models (LLMs) capable of natural language reasoning (Zawacki-Richter et al., 2019). Traditional academic decision-support tools relied on static `if-then-else` decision trees, which lacked conversational flexibility and could not handle nuanced student inquiries regarding career scope, interdisciplinary options, or localized admission procedures (Wollny et al., 2021).

Recent breakthroughs in generative transformer architectures—exemplified by Google Gemini, OpenAI GPT, and Anthropic Claude—have demonstrated remarkable proficiency in synthesized educational advising (Kasneci et al., 2023). LLMs can contextualize student academic records, explain complex admission formulas in plain language, and tailor career advice to student interests. However, scholars have raised critical concerns regarding the deployment of unconstrained generative models in high-stakes domains, citing risks of algorithmic hallucination, fabrication of nonexistent admission criteria, and failure during network interruptions (Bender et al., 2021).

To mitigate these risks, recent literature advocates for **grounded hybrid advising architectures** (Marrone et al., 2022). In such architectures, generative models are constrained by prompt-engineered domain rules, supplied with factual context extracted from trusted local databases, and backed by deterministic rule-based fallback engines. EduVerse AI implements precisely this resilient paradigm by combining Google Gemini API with a deterministic rule-based counseling engine.

2.4 College Search and Comparison Platforms

Comparative evaluation constitutes an essential cognitive phase in consumer and educational decision-making. According to human-computer interaction (HCI) research by Tversky (1972) and later formalized by Pu et al. (2011), users evaluating complex multi-attribute alternatives experience severe cognitive overload when forced to switch between separate browser tabs to compare institutional parameters.

Visual comparison matrices that align candidate entities across standardized attribute rows significantly improve decision accuracy and reduce cognitive fatigue (Spence, 2007). In higher education platforms, the primary challenge in generating comparison matrices lies in data heterogeneity: different colleges report tuition fees on varying scales (per semester vs. per annum), describe accreditation differently, and offer courses under non-standardized nomenclatures. Effective platforms must normalize institutional records into a canonical schema prior to presentation. EduVerse AI achieves this through a centralized relational model and an interactive comparison engine supporting side-by-side evaluation of up to four institutions.

2.5 Web-Based Information Retrieval

Information retrieval (IR) within localized educational domains requires balancing high recall with precise domain filtering (Manning, Raghavan, & Schütze, 2008). In the context of Tamil Nadu collegiate discovery, standard keyword search frequently encounters geographic and naming ambiguities. For example, institutional names often contain overlapping terms such as "College of

Engineering", "Institute of Technology", or references to district headquarters versus surrounding taluks.

Effective domain-specific information retrieval necessitates combining database-level string matching (SQL `LIKE` / `ILIKE`) with multi-attribute faceted indexing (Hearst, 2006). Faceted search allows students to progressively narrow candidate result sets by applying orthogonal categorical filters (district, college type, autonomous status, fee ceiling) without losing search context. EduVerse AI implements faceted search using parameterized SQLAlchemy queries, ensuring sub-second response times across large result sets.

2.6 Research Gaps and Project Motivation

A synthesis of existing academic literature and practical industry platforms reveals several critical research and implementation gaps:

Identified Research / Implementation Gap	Existing Commercial / Academic Status	EduVerse AI Architectural Solution
Commercial Placement Bias	Commercial portals prioritize institutions based on advertising fees and sponsored bids.	Fully objective, merit- and preference-based scoring without sponsored listings.
Data Fragmentation across Tamil Nadu Districts	Portals focus primarily on major tier-1 cities (Chennai, Coimbatore), neglecting rural districts.	Comprehensive coverage of all 38 districts with 1,129 verified colleges.
Unverified and Broken Official URLs	Aggregators list broken URLs or route traffic to third-party telemarketing forms.	100% of indexed colleges mapped to verified official web URLs opening securely in new tabs.
Opaque Recommendation Logic	Platforms offer "black-box" suggestions without explaining why a college was recommended.	Transparent multi-criteria scoring algorithm displaying explainable match factors.
Disconnection between Cutoffs and Recommendations	Cutoff calculation and college matching are treated as disconnected tools on separate sites.	Integrated assessment pipeline converting 12th marks into TNEA cutoffs and ranked matching.
Vulnerability to AI Hallucinations & API Outages	Standalone AI chatbots hallucinate admission criteria and fail during internet or API quota limits.	Grounded hybrid architecture coupling Google Gemini with an autonomous rule-based fallback counselor.

Addressing these gaps motivates the design, implementation, and academic evaluation of **EduVerse AI**.

CHAPTER 3 – SYSTEM ANALYSIS

3.1 Existing System

In the existing ecosystem, students concluding their higher secondary education in Tamil Nadu rely on a fragmented collection of resources to explore colleges and undergraduate programs. These resources include:

- **Printed Admission Booklets:** Published annually during state counseling rounds, these booklets contain dense tabular listings of colleges, sanctioned intakes, and historical cutoff ranges. However, they are static, cannot be filtered interactively, lack search functionality, and provide no visual comparative insights.

- **State Counseling Web Portals (e.g., TNEA, TNGASA):** While authoritative, official government counseling portals operate primarily during active admission cycles. Their interfaces prioritize regulatory administrative functions rather than student discovery, career guidance, or comprehensive institutional profiling.

- **Commercial Aggregator Websites:** Commercial portals maintain extensive directories but suffer from sponsored ranking distortions, aggressive pop-up advertisements, lead-generation paywalls, and unverified hyperlinks.

- **Word-of-Mouth and Informal Counseling:** Rural and first-generation students heavily depend on advice from family members, school teachers, or local coaching centers. While well-intentioned, informal guidance is inherently constrained by the advisor's limited awareness of modern interdisciplinary specializations and statewide institutional options.

Limitations of the Existing System:

1. Absence of a centralized, student-centric, non-commercial digital platform for Tamil Nadu colleges.

2. Inability to perform side-by-side comparative analysis of candidate colleges across standardized metrics.

3. Manual, error-prone calculation of TNEA cutoffs without automated course eligibility matching.

4. Total absence of explainable recommendation algorithms tailored to student financial ceilings and campus preferences.

5. Inability to interact with an intelligent, 24/7 conversational educational advisor without paying consultation fees.

3.2 Proposed System

The proposed **EduVerse AI** system is a web-based, AI-augmented academic decision-support platform engineered specifically to resolve the deficiencies of existing approaches. Designed as an MCA full-stack solution, EduVerse AI combines structured relational data management with modern client-side interactivity, content-based recommendation algorithms, and conversational AI services.

Key Innovations of the Proposed System:

- **Statewide Institutional Catalog:** Indexes 1,129 colleges across all 38 districts of Tamil Nadu, categorized by a multi-dimensional taxonomy (Government, Government-Aided, Self-Financing, Autonomous status) and enriched with verified official website URLs.

- **Dynamic Multi-Faceted Discovery:** Empowers students to search and filter colleges by district, affiliation, accreditation, course offering, and budget ceiling with immediate UI feedback.

- **Interactive Multi-College Comparison:** Provides a dedicated comparison matrix allowing students to analyze up to four institutions simultaneously across fee structures, ratings, campus facilities, and geographical coordinates.

- **Automated Academic Assessment Engine:** Ingests 12th standard subject marks, computes aggregate percentages, and calculates official TNEA cutoffs on both standardized 200-mark and formula scales.

- **Explainable Recommendation Service:** Employs a multi-criteria scoring algorithm (evaluating course match 35%, location affinity 25%, budget ceiling 20%, college type 10%, and campus facilities 10%) that outputs ranked recommendations with human-readable match justifications.

- **Hybrid Conversational AI Counselor:** Combines the natural language understanding of Google Gemini API with a deterministic educational rule-based fallback engine, ensuring uninterrupted academic guidance.

- **Live Educational Analytics:** Renders dynamic data visualizations depicting district-wise institutional concentrations, course category distributions, and management types using Chart.js.

- **Secure Administrative Governance:** Provides administrative controllers with full CRUD capabilities over colleges, courses, facilities, student inquiries, and system audit logs.

3.3 Functional Requirements

The functional requirements describe the specific operations and behaviors that EduVerse AI must perform:

1. **User Authentication and Session Management:**

- The system shall allow students to register using their name, email address, and a secure password.

- The system shall authenticate users and maintain encrypted, HTTP-only session cookies.

- The system shall enforce role-based access control distinguishing Students from Administrators.

2. **College Discovery and Filtering:**

- The system shall enable keyword search across college names, cities, districts, and institutional codes.

- The system shall provide faceted filtering by district (all 38 districts), college type, autonomous status, course discipline, and tuition fee range.

- The system shall paginate search results (9 colleges per page) with navigation controls.

3. **College Profiling and Official Web Navigation:**

- The system shall render a detailed institutional profile displaying address, affiliation, accreditation, courses, annual fees, and amenities.

- The system shall display a "Visit Official Website" button that routes users directly to the verified institutional domain in a new tab with `target="_blank"` and `rel="noopener noreferrer"`.

- The system shall display verified Google Maps links and driving direction URLs for institutions with valid geographic coordinates.

4. **Institutional Comparison:**

- The system shall enable students to add candidate colleges to a temporary comparison tray stored in session memory.

- The system shall render a tabular comparison matrix evaluating selected colleges side-by-side across courses, fees, accreditations, and amenities.

5. **Academic Marks Assessment and Cutoff Computation:**

- The system shall accept 12th standard marks across academic streams (Science - Maths, Science - Bio, Commerce, Arts).

- The system shall compute overall percentage and standard TNEA cutoff:

$$\text{TNEA Cutoff} = \frac{\text{Maths}}{2} + \frac{\text{Physics}}{4} + \frac{\text{Chemistry}}{4}$$

6. **Intelligent Course and College Recommendation:**

- The system shall ingest student preferences (desired stream, location, budget ceiling, campus facilities).

- The system shall compute normalized match scores (0 to 100) and generate human-readable match explanations.

7. **Conversational AI Educational Advising:**

- The system shall process student natural language queries through Google Gemini API.

- The system shall fall back automatically to an internal rule-based guidance engine if the API key is unconfigured or network requests fail.

8. **Administrative Catalog Governance:**

- The system shall allow authorized administrators to create, update, and soft-delete colleges and courses.

- The system shall provide administrative review workflows for student queries and system feedback.

3.4 Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints governing system execution:

1. **Performance and Latency:** The system shall achieve a server response time of less than 200 milliseconds for catalog search and filtering under standard local server loads. Dynamic API endpoints shall deliver JSON payloads in under 100 milliseconds.

2. **Usability and Aesthetics:** The user interface shall follow a modern, cohesive SaaS aesthetic utilizing a professional educational blue color palette (`#1E3A8A`, `#2563EB`, `#EFF6FF`, `#F8FAFC`, `#14B8A6`). The layout shall maintain consistent vertical rhythm, equal card heights, and clear typographic hierarchy.

3. **Reliability and Availability:** The system shall guarantee continuous operation by deploying an autonomous rule-based fallback engine that ensures counseling functionality remains operational even during third-party AI outages.

4. **Maintainability and Modularity:** The codebase shall strictly adhere to the Flask Application Factory pattern, separating routing controllers (Blueprints), business logic (Services), and database interactions (Models).

5. **Security and Data Protection:** The system shall protect all web forms against Cross-Site Request Forgery (CSRF) via Flask-WTF tokens. Database queries shall use parameterized SQLAlchemy ORM statements to eliminate SQL injection risks. Passwords shall be hashed using PBKDF2 with SHA-256 salts.

6. **Cross-Platform Responsiveness:** The web application shall render fluidly across desktop monitors, laptop screens, tablets, and mobile smartphones through CSS Grid and Flexbox layouts.

7. **Accessibility (a11y):** Form fields shall include explicit associated `<label>` tags, interactive buttons shall feature discernible focus rings, and text elements shall satisfy WCAG AA contrast ratio thresholds (> 4.5:1).

3.5 Hardware Requirements

The system operates efficiently on commodity personal computing hardware for local deployment and development:

Table 3.1: Minimum and Recommended Hardware Specifications

Hardware Component	Minimum Development Requirement	Recommended Production Specification
Central Processing Unit (CPU)	Dual-Core Intel Core i3 / AMD Ryzen 3 (2.0 GHz)	Quad-Core Intel Core i5 / AMD Ryzen 5 or higher
System Memory (RAM)	4 GB DDR4	8 GB or 16 GB DDR4/DDR5
Secondary Storage	20 GB available disk space (HDD)	50 GB Solid State Drive (NVMe SSD)
Network Interface	Standard Broadband (1 Mbps connection)	High-Speed Broadband / Ethernet (100 Mbps)
Display Resolution	1366 × 768 pixels	1920 × 1080 pixels (Full HD)

3.6 Software Requirements

The software dependencies and development environments used across EduVerse AI are specified below:

Table 3.2: Software Environment and Framework Dependencies

Software / Package	Version	Functional Purpose	Source File Where Employed
Python	3.10.11 / 3.11+	Core runtime language environment	Entire project ecosystem
Flask	3.1.0	Web framework and routing engine	`app/__init__.py`, `run.py`
Flask-SQLAlchemy	3.1.1	Object-Relational Mapping (ORM)	`app/models/`, `app/__init__.py`
SQLAlchemy	2.0.32	Database abstraction and query engine	`app/services/college_service.py`

Flask-Migrate	4.1.0	Database schema migrations via Alembic	`app/__init__.py`
Flask-WTF	1.2.2	CSRF protection and form handling	`app/__init__.py`, `app/forms/`
email-validator	2.2.0	Syntactic validation of user email addresses	`app/utils/validators.py`
PyMySQL	1.1.1	Pure-Python MySQL database driver	`config.py`
cryptography	44.0.2	Cryptographic primitives for database access	`config.py`
pandas	2.2.3	Tabular feature processing for recommender	`app/recommender/`
numpy	2.2.3	Numerical array operations and vector math	`app/recommender/`
scikit-learn	1.6.1	Feature encoding and distance calculations	`app/recommender/`
python-dotenv	1.0.1	Configuration management via `.env`	`config.py`
pytest	8.3.5	Automated test runner and assertion framework	`tests/`
Jinja2	3.1.5	Server-side HTML template rendering engine	`app/templates/`
Werkzeug	3.1.3	WSGI utilities and password security	`app/models/user.py`
Chart.js	4.4.0 (CDN)	Interactive client-side data visualization	`app/templates/analytics.html`
Font Awesome	6.4.0 (CDN)	Standardized user interface iconography	`app/templates/base.html`

3.7 Feasibility Study

Prior to development, a feasibility study was conducted to evaluate the viability of EduVerse AI across four dimensions:

1. **Technical Feasibility:** Python Flask is a mature, lightweight, and extensible web framework. Paired with SQLAlchemy, it supports seamless transitions between SQLite (development) and MySQL/PostgreSQL (production). Client-side requirements are satisfied using vanilla modern JavaScript and Chart.js, eliminating heavy frontend build steps. The integration of the Google Gemini

REST API requires only standard HTTP client libraries (`urllib.request`), while the rule-based fallback guarantees system availability. The technical architecture is fully feasible.

2. **Operational Feasibility:** The system is engineered specifically for 12th standard students and parents who possess basic digital literacy. The intuitive search interface, clear comparison matrices, and step-by-step academic assessment workflows require no specialized training. Administrative workflows mirror standard content management systems. Operationally, the platform provides immediate practical utility.

3. **Economic Feasibility:** The entire software stack comprises open-source software libraries licensed under permissive terms (BSD, MIT, Apache 2.0). The database runs locally on SQLite or free tiers of MySQL. The Google Gemini API operates within free-tier quotas for educational applications, and the rule-based fallback incurs zero ongoing API costs. Thus, development and operational expenditures are minimal.

4. **Schedule Feasibility:** The project was planned and executed within the designated semester schedule. Modular decomposition into Blueprints (Auth, Colleges, Comparison, Recommendations, Analytics, Admin, Student) facilitated parallel development, automated testing, and incremental refinement.

3.8 System Constraints

The operational boundaries and constraints of the current EduVerse AI implementation include:

- **Local Application Server:** In the current prototype configuration, the Flask application runs on a local WSGI server (`http://127.0.0.1:5000`) without an enterprise reverse proxy (e.g., Nginx, Gunicorn).

- **Curated Institutional Dataset:** The system indexes 1,129 colleges across Tamil Nadu. While extensive, specialized standalone polytechnic or vocational diplomas are outside the catalog's scope.

- **Third-Party AI API Rate Limits:** Live generative counseling depends on external Google Gemini API availability. To operate within free-tier rate limits, responses are capped at 650 tokens, and requests fall back to internal rules when quotas are exceeded.

- **Static Geolocation Mapping:** Institutional latitude and longitude coordinates are indexed in the database and linked to Google Maps, but real-time GPS geolocation of user devices is not automatically tracked.

CHAPTER 4 – SYSTEM DESIGN

4.1 Overall System Architecture

EduVerse AI is designed around a multi-tier, modular web application architecture. The system establishes a clean separation of concerns among the presentation tier, client-side scripts, routing controllers, business services, recommendation models, and relational data stores.

Figure 4.1 depicts the overall architectural topology of the platform:



```
|   +-------------------------------------------------------------------------------+   |
|   | Relational Store: SQLite (instance/eduverse_dev.db) / MySQL (via PyMySQL)     |   |
+-------------------------------------------------------------------------------------+
```

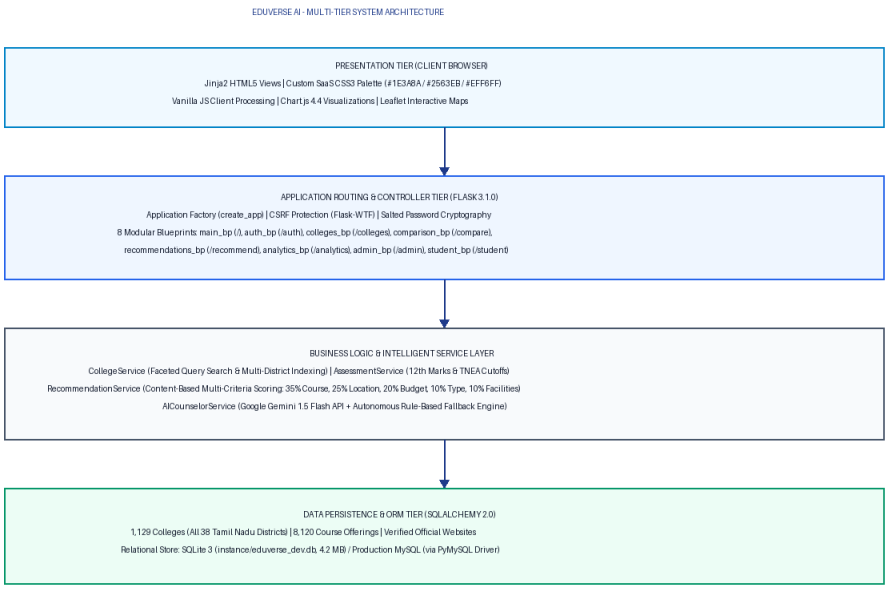


Figure 4.1: Overall System Architecture of EduVerse AI

4.2 Layered Architectural Design

The architecture is structured into cohesive functional layers:

4.2.1 Presentation Layer

Renders server-side HTML5 views using the Jinja2 templating engine. The design relies on a unified master layout (`base.html`) containing common navigation headers, user profile dropdowns, flash message containers, and footer links. Sub-views extend this base layout to present specific modules: landing page, college directory, comparison matrix, assessment forms, analytics dashboards, and administrative tables.

4.2.2 Client-Side Processing Layer

Implemented in vanilla JavaScript (`main.js`, `compare.js`, `analytics.js`). Handles asynchronous `fetch()` requests for dynamic filtering, comparison list manipulation, Chart.js canvas rendering, and modal dialog lifecycle management without page reloads.

4.2.3 Flask Application Layer

The core WSGI application initialized via `create_app()` in `app/__init__.py`. Manages application configuration, extension binding (SQLAlchemy, Migrate, CSRFProtect), HTTP error handlers (404, 500, 403), request-lifecycle hooks (`@before_request`), and Jinja context processors (`inject_globals`).

4.2.4 API and Routing Layer

Organized across 8 Flask Blueprints encapsulating 66 registered routes. Exposes both HTML-rendering endpoints and RESTful JSON endpoints for AJAX-driven interactions.

4.2.5 Business Logic Layer

Encapsulates domain-specific rules within stateless Python service classes. Isolates algorithmic workflows (TNEA cutoff formula computation, recommendation scoring, query triage) from route handling.

4.2.6 Data Access Layer

Leverages SQLAlchemy ORM to map Python classes to relational database tables. Enforces database constraints, foreign-key relationships, index definitions, and cascades while protecting against SQL injection via parameterized queries.

4.2.7 AI and Recommendation Layer

Combines machine learning feature engineering (`CollegeFeatureExtractor`), multi-attribute scoring (`RecommendationScorer`), and generative conversational advising (`AICounselorService`).

4.2.8 Analytics Layer

Executes real-time aggregate SQL queries (e.g., `func.count`, `func.group_by`) across 1,129 colleges to compile district distributions, accreditation breakdowns, and course availability for visualization.

4.3 Use Case Diagram

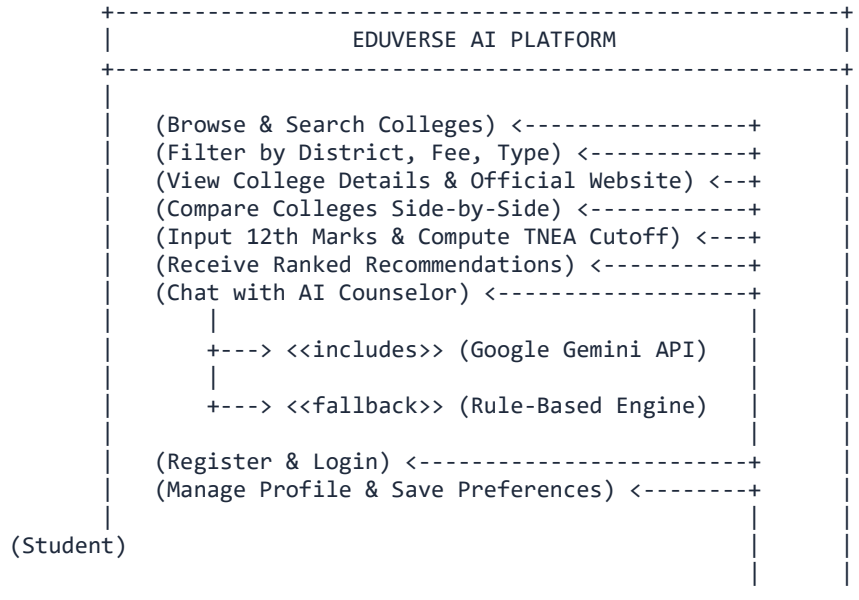
The primary actors interacting with EduVerse AI are:

1. **Student / Guest User:** Searches colleges, filters by district and course, views verified profiles, visits official websites, calculates cutoffs, compares institutions, receives recommendations, and chats with the AI counselor.

2. **Authenticated Student:** Maintains personal academic profile, saves target preferences, tracks assessment history, and submits inquiries.

3. **Administrator:** Authenticates into the protected admin portal, performs CRUD operations on colleges and courses, monitors analytics, and resolves student queries.

4. **External AI Service (Google Gemini API):** Ingests contextualized student prompts and returns structured educational guidance.



```
(Administrator) ----------------------------------------+      |
       |                                                       |
       |   (Manage Colleges - CRUD)                            |
       |   (Manage Courses & Facilities)                       |
       |   (Review Student Queries & Feedback)                 |
       |   (View System Analytics & Login Logs)                |
      +-------------------------------------------------------+
```



Figure 4.3: UML Use Case Diagram of EduVerse AI Platform

4.4 Data Flow Diagrams

Context Level Data Flow Diagram (DFD Level-0)

The Context Diagram illustrates the macro-level boundary of EduVerse AI, identifying external entities and primary information exchanges:

```
[Student User]
       |
       |--- 1. Search Criteria, 12th Marks, Preferences, Chat Queries --->
       |<-- 2. Filtered Colleges, Official Links, Cutoffs, Advice --------
       |
       v
+=========================================================================+
|                        0.0 EDUVERSE AI SYSTEM                           |
|      (College Discovery, Comparison, Cutoff & AI Guidance)              |
+=========================================================================+
       ^                                    ^                    ^
       |                                    |                    |
       |--- 3. Credentials, Updates --->    |--- 5. Prompts --->  |
       |<-- 4. Audit Data, Logs --------    |<-- 6. Answers ----  |
       |                                    |                    |
[Administrator]                    [Google Gemini API]    [(Database Store)]
```

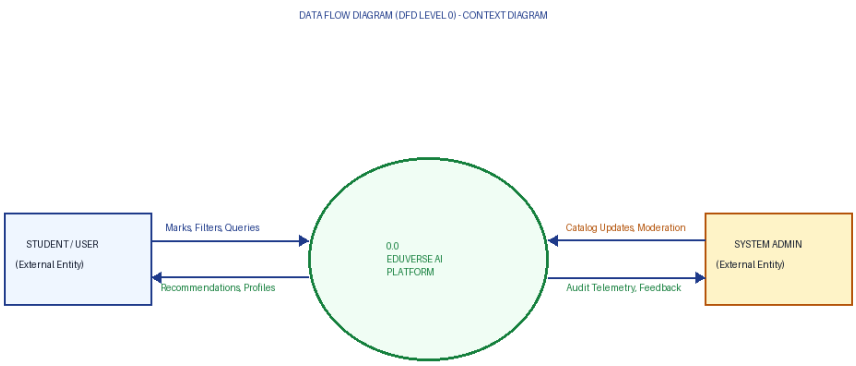


Figure 4.4: Context Level Data Flow Diagram (DFD Level-0)

Level-1 Data Flow Diagram

The Level-1 DFD decomposes the system into core operational sub-processes:

```
[Student] ---> (1.0 Ingest Marks & Stream) ---> [StudentProfile Store]
                          |
                          v
               (2.0 Calculate Percentage & TNEA Cutoff)
                          |
                          v (Calculated Cutoff & Profile Vector)
[Preferences] -> (3.0 Multi-Criteria Scoring & Ranking) <--- [Colleges Store]
                          |
                          v
               (4.0 Generate Ranked Recommendations & Factors)
                          |
                          +---> Display Recommendations on Dashboard
                          |
                          v (Context Vector)
[Chat Query] --> (5.0 AI Counselor Triage)
                          |
                          |---> [Google Gemini API] (Primary)
                          |           | (Response)
                          |           v
                          +---> (6.0 Format & Render Guidance) ---> [Student]
                          |           ^
                          | (Fallback)|
                          +---> [Rule-Based Engine]
```



Figure 4.5: Level-1 Data Flow Diagram for Student Assessment and Recommendation

4.5 Application Workflow

The end-to-end user journey follows an intuitive multi-step sequence:

1. **Entry and Discovery:** The student accesses the landing page (`/`), views statewide district aggregations, and initiates an institutional search (`/colleges`).

2. **Faceted Filtering:** The student applies filters (e.g., Coimbatore district, Engineering courses, Autonomous status, budget ceiling). The server filters the database and returns matched college cards.

3. **Official Website Verification:** On the college details page (`/colleges/<id>`), the student reviews courses, facilities, accreditation, and clicks **"Visit Official Website"**, which launches the verified institutional URL in a secure new browser tab.

4. **Side-by-Side Comparison:** The student adds candidate colleges to their comparison tray and navigates to `/compare` to examine differences in tuition, ratings, and course offerings.

5. **Academic Assessment:** The student accesses the assessment module (`/student/assessment`), selects their 12th standard stream, enters subject marks, and clicks "Calculate & Recommend".

6. **Cutoff and Recommendation Output:** The backend calculates the aggregate percentage and official TNEA cutoff, executes the multi-criteria recommendation scoring model, and renders ranked institutional matches with transparent match factors.

7. **Interactive AI Advising:** The student opens the AI Counselor (`/student/ai-counselor`), submits questions regarding course choices or career options, and receives personalized answers.

4.6 Database Design

EduVerse AI uses a normalized relational database schema managed via SQLAlchemy ORM. The schema consists of 12 structured tables:

4.6.1 Table Specifications

Table 4.1: Database Entity Specification: `colleges` Table

Column Name	Data Type	Constraints	Description
`id`	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique college identifier
`name`	VARCHAR(255)	NOT NULL, INDEXED	Full institutional name
`code`	VARCHAR(50)	NULLABLE, INDEXED	TNEA / AISHE regulatory institutional code
`state`	VARCHAR(100)	NOT NULL, DEFAULT 'Tamil Nadu'	State location
`district_id`	INTEGER	FOREIGN KEY (`districts.id`), INDEXED	Reference to administrative district
`district`	VARCHAR(100)	NOT NULL, INDEXED	District name for direct query speed
`city`	VARCHAR(100)	NOT NULL, INDEXED	City or municipal corporation
`college_type`	VARCHAR(100)	NOT NULL, INDEXED	Government, Aided, Self-Financing, etc.
`primary_college_type`	VARCHAR(50)	NOT NULL, DEFAULT 'UNKNOWN'	Multi-dimensional taxonomy classification
`autonomous_status`	VARCHAR(50)	NOT NULL, DEFAULT 'NON_AUTONOMOUS'	Autonomous or Non-Autonomous status
`normalized_name`	VARCHAR(255)	NULLABLE, INDEXED	Cleaned lowercase institutional name
`official_website`	VARCHAR(500)	NULLABLE	Verified official institutional website URL
`official_website_url`	VARCHAR(500)	NULLABLE	Canonical verified website URL mirror
`website_status`	VARCHAR(50)	NOT NULL, DEFAULT 'UNVERIFIED'	VERIFIED, UNVERIFIED, MISSING, BROKEN
`latitude`	FLOAT	NULLABLE	Geographic latitude coordinate
`longitude`	FLOAT	NULLABLE	Geographic longitude coordinate
`accreditation`	VARCHAR(100)	NULLABLE	NAAC rating (e.g., 'NAAC A++', 'NAAC A+')

`affiliation`	VARCHAR(150)	NULLABLE	University affiliation (e.g., 'Anna University')
`rating`	FLOAT	NULLABLE, DEFAULT 4.2	Composite institutional rating (1.0 to 5.0)
`data_status`	VARCHAR(50)	NOT NULL, DEFAULT 'SAMPLE'	VERIFIED, UNVERIFIED, SAMPLE
`created_at`	DATETIME	NOT NULL	Creation timestamp (UTC)
`last_updated`	DATETIME	NOT NULL	Last modification timestamp (UTC)

Table 4.2: Database Entity Specification: `courses` Table

Column Name	Data Type	Constraints	Description
`id`	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique course identifier
`college_id`	INTEGER	FOREIGN KEY (`colleges.id`), NOT NULL	Owning college reference
`name`	VARCHAR(150)	NOT NULL, INDEXED	Course name (e.g., 'B.E. Computer Science')
`duration_years`	INTEGER	NOT NULL, DEFAULT 4	Degree duration in years
`annual_fee`	FLOAT	NULLABLE, INDEXED	Annual tuition fee in Indian Rupees (INR)
`sanctioned_intake`	INTEGER	NULLABLE	Number of approved enrollment seats
`cutoff_marks`	FLOAT	NULLABLE, INDEXED	Historical entry cutoff mark
`stream`	VARCHAR(50)	NULLABLE	Engineering, Arts, Science, Commerce

Table 4.3: Database Entity Specification: `districts` Table

Column Name	Data Type	Constraints	Description
`id`	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique district identifier
`name`	VARCHAR(100)	NOT NULL, UNIQUE, INDEXED	Standardized Tamil Nadu district name
`code`	VARCHAR(10)	NULLABLE, UNIQUE	Official administrative district code

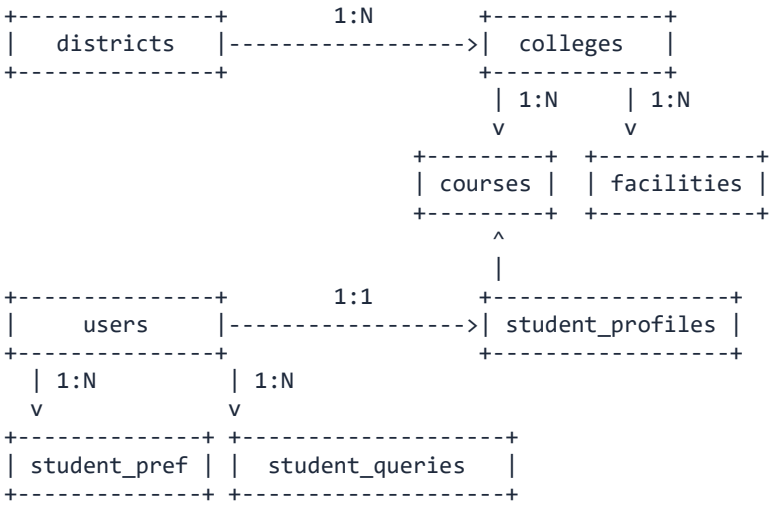
`zone`	VARCHAR(50)	NULLABLE	Geographic zone (North, South, West, Central)

Table 4.4: Database Entity Specification: `users` Table

Column Name	Data Type	Constraints	Description
`id`	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique user identifier
`name`	VARCHAR(150)	NOT NULL	Full name of registered user
`email`	VARCHAR(150)	NOT NULL, UNIQUE, INDEXED	Unique email address
`password_hash`	VARCHAR(255)	NOT NULL	Salted PBKDF2-SHA256 password hash
`role`	VARCHAR(20)	NOT NULL, DEFAULT 'student'	Role: 'student' or 'admin'
`created_at`	DATETIME	NOT NULL	Account registration timestamp

4.6.2 Entity-Relationship (ER) Model

The relational topology is anchored by the `colleges` table, which maintains one-to-many relationships with `courses` and `facilities`, a many-to-one relationship with `districts`, and a one-to-many relationship with `comparison_items`. Registered `users` maintain a one-to-one relationship with `student_profiles` and one-to-many relationships with `student_preferences`, `student_queries`, and `student_feedbacks`.



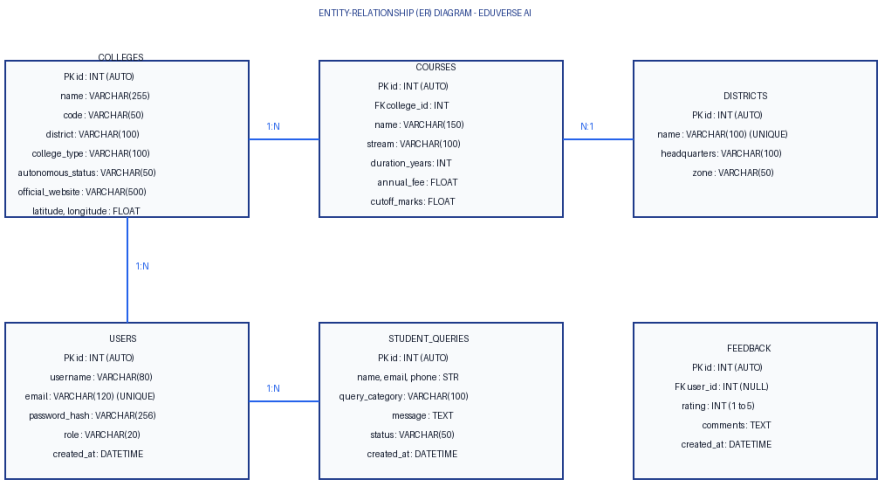


Figure 4.7: Entity-Relationship (ER) Diagram of EduVerse AI Database

4.7 API Design

The platform exposes 66 routes across 8 Blueprints. The principal web and JSON API routes are documented below:

Table 4.6: Comprehensive API Endpoint Catalog

HTTP Method	Route Endpoint	Blueprint Module	Handler Function	Request Parameters / Payload	Response Type	Purpose / Description
`GET`	`/`	`main`	`index()`	None	HTML	Landing page with district counts and search entry
`GET`	`/about`	`main`	`about()`	None	HTML	About page with architecture and MCA student credits
`GET`	`/colleges/`	`colleges`	`colleges_page()`	`q`, `district`, `type`, `page`	HTML	Paginated college search and catalog view
`GET`	`/colleges/<id>`	`colleges`	`college_detail()`	`id` (path)	HTML	Detailed institutional profile with official website

`GET`	`/colleges/api/districts`	`colleges`	`api_districts()`	None	JSON	Returns list of all 38 Tamil Nadu districts
`GET`	`/compare`	`comparison`	`compare_view()`	None	HTML	Side-by-side college comparison matrix
`POST`	`/compare/api/add`	`comparison`	`add_to_compare()`	`{"college_id": int}`	JSON	Adds college to session comparison tray
`POST`	`/compare/api/remove`	`comparison`	`remove_from_compare()`	`{"college_id": int}`	JSON	Removes college from comparison tray
`GET`	`/recommend`	`recommendations`	`recommend_page()`	None	HTML	Multi-attribute recommendation input form
`POST`	`/recommend/api`	`recommendations`	`api_recommend()`	`{"district", "course", "budget"}`	JSON	Content-based ranked recommendations
`GET`	`/analytics`	`analytics`	`analytics_page()`	None	HTML	Interactive analytics dashboard view
`GET`	`/api/analytics`	`analytics`	`api_analytics()`	None	JSON	District and course distribution metrics for Chart.js
`GET`	`/student/dashboard`	`student`	`dashboard()`	None	HTML	Student guidance overview and assessment summary
`POST`	`/student/api/marks-evaluate`	`student`	`api_marks_evaluate()`	`{"stream", "marks": {}}`	JSON	Calculates aggregate percentage and TNEA cutoff
`POST`	`/student/api/ai-advisor`	`student`	`api_ai_advisor()`	`{"message": str}`	JSON	Generates AI counselor

						guidance response
`GET`	`/admin`	`admin`	`dashboard()`	`page`, `q`	HTML	Admin dashboard with college list and query triage
`POST`	`/admin/colleges/new`	`admin`	`new_college()`	Form data	Redirect / HTML	Creates new college record in database
`POST`	`/admin/colleges/<id>/edit`	`admin`	`edit_college()`	Form data	Redirect / HTML	Updates existing college record

4.8 User Interface Design

The user interface is designed to provide an uncluttered, modern, and student-friendly experience:

- **Design System and Color Palette:** Implements a professional SaaS educational theme featuring deep primary blue (`#1E3A8A`) for branding and major headers, interactive secondary blue (`#2563EB`) for action buttons, soft blue tint (`#EFF6FF`) for highlight sections, crisp white (`#FFFFFF`) for card containers, subtle gray (`#E2E8F0`) for borders, and teal (`#14B8A6`) for success indicators.

- **Equal-Height Flexbox Layouts:** Card grids use Flexbox (`display: flex; flex-direction: column; height: 100%;`) with `flex-grow: 1` applied to content areas, ensuring uniform card heights and bottom-aligned action buttons across varying text lengths.

- **Verified Official Website Action Button:** Institutional profile cards render a prominent, styled button labeled **"Visit Official Website"**. This button renders exclusively when a valid, sanitized URL exists, opening the verified institutional domain in a new tab with security attributes (`target="_blank" rel="noopener noreferrer"`).

- **Responsive Mobile Navigation:** Navigation bars collapse into a clean, accessible mobile drawer menu on viewport widths below 768 pixels.

4.9 Design Principles and Architectural Advantages

The technical implementation follows fundamental software engineering principles:

1. **Modularity:** Each functional capability resides within a self-contained Blueprint and Service class, enabling independent testing and maintenance.

2. **Separation of Concerns:** Business logic (TNEA cutoff math, scoring algorithms) is isolated from HTTP controllers and presentation templates.

3. **Defense in Depth:** Input validation occurs at the client layer, route validation layer, and database ORM layer.

4. **Resilience via Graceful Degradation:** The AI Counselor operates with an autonomous rule-based fallback, ensuring unbroken service delivery during API quota exhaustion or network downtime.

CHAPTER 5 – SYSTEM IMPLEMENTATION

5.1 Development Environment

The implementation of EduVerse AI was executed in a Python virtual environment running on Windows 11:

- **Runtime Environment:** Python 3.10.11 Virtual Environment (`.\venv\Scripts\python.exe`)

- **Web Application Framework:** Flask 3.1.0 with Jinja2 3.1.5

- **ORM & Data Layer:** SQLAlchemy 2.0.32 and Flask-SQLAlchemy 3.1.1

- **Database Engine:** SQLite 3 (primary local database: `instance/eduverse_dev.db`, 4.2 MB) with production PyMySQL 1.1.1 drivers

- **Integrated Development Environment (IDE):** Visual Studio Code with Python, Pylance, and SQLite Viewer extensions

- **Testing Framework:** pytest 8.3.5

5.2 Project Directory Structure

The codebase follows a standardized enterprise Flask project structure:

```
c:\Users\User1\Documents\eduverseAI (new)\
|-- run.py                                  # Application entry point and server startup
|-- config.py                               # Environment configurations (Dev, Test, Prod)
|-- requirements.txt                        # Pinned Python package dependencies
|-- generate_doc_report.py                  # Report compilation utility
|-- instance/
|   `-- eduverse_dev.db                     # SQLite database (4.2 MB, 1,129 colleges)
|-- app/
|   |-- __init__.py                         # Application Factory (create_app)
|   |-- models/                             # SQLAlchemy ORM entity models
|   |   |-- __init__.py                     # Package export manifest
|   |   |-- college.py                      # College entity with URL sanitization
|   |   |-- course.py                       # Course entity with fee and cutoff attributes
|   |   |-- district.py                     # District entity for Tamil Nadu's 38
districts
|   |   |-- facility.py                     # Campus amenities entity
|   |   |-- user.py                         # User entity with salted password hashing
|   |   |-- student_profile.py              # Extended student academic profile
|   |   |-- student_query.py                # Student inquiry entity for admin triage
|   |   `-- preference.py                   # Saved student search preferences
|   |-- routes/                             # Flask Blueprints (Controllers)
|   |   |-- main.py                         # Landing, About, District Explorer routes
|   |   |-- auth.py                         # Login, Registration, Logout routes
|   |   |-- colleges.py                     # College search, filter, and detail routes
|   |   |-- comparison.py                   # College comparison matrix routes
|   |   |-- recommendations.py              # Multi-attribute recommendation routes
|   |   |-- analytics.py                    # Real-time analytics view & JSON routes
|   |   |-- admin.py                        # Administrator CRUD and governance routes
|   |   `-- student.py                      # Assessment, Cutoff math, AI Counselor routes
|   |-- services/                           # Stateless business logic services
|   |   |-- college_service.py              # Faceted query building and district
retrieval
|   |   |-- assessment_service.py           # Percentage and TNEA cutoff calculation
|   |   |-- recommendation_service.py       # Recommendation workflow coordinator
|   |   |-- analytics_service.py            # Aggregate database analytics calculator
|   |   `-- ai_counselor_service.py         # Google Gemini & rule-based guidance engine
|   |-- recommender/                        # Recommendation algorithmic pipeline
```

```
|   |   |-- recommender.py                # ContentBasedRecommender pipeline
|   |   |-- scoring.py                    # RecommendationScorer (weights & factors)
|   |   `-- feature_engineering.py        # Dataframe feature extraction
|   |-- utils/                            # Shared utility helpers and decorators
|   |   |-- validators.py                 # Input validation routines
|   |   `-- decorators.py                 # @admin_required security decorators
|   |-- static/                           # Web static assets
|   |   |-- css/
|   |   |   `-- styles.css                # Custom SaaS educational stylesheet
|   |   `-- js/
|   |       |-- main.js                   # Global UI interactions and search debouncing
|   |       |-- compare.js                # Comparison tray and matrix manipulation
|   |       `-- analytics.js              # Chart.js visualization renderer
|   `-- templates/                        # Jinja2 HTML templates
|       |-- base.html                     # Master HTML5 skeleton with header/footer
|       |-- index.html                    # Landing page with district counters
|       |-- about.html                    # System mission and MCA academic credits
|       |-- analytics.html                # Interactive Chart.js analytics dashboard
|       |-- colleges/                     # College catalog and comparison views
|       |-- student/                      # Assessment, results, and AI counselor views
|       |-- admin/                        # Administrative dashboard and forms
|       `-- errors/                       # Custom 404, 500, 403 error pages
`-- tests/                                # Automated test suite (pytest)
    |-- conftest.py                       # Shared test fixtures (in-memory SQLite)
    |-- test_admin.py                     # Admin controller and security tests
    |-- test_analytics.py                 # Analytics calculation tests
    |-- test_auth.py                      # User authentication and session tests
    |-- test_college_service.py           # Faceted search and service tests
    `-- test_recommendations.py           # TNEA cutoff and recommendation tests
```

5.3 Flask Application Initialisation

The application initializes via `create_app()` in `app/__init__.py`. It reads configuration objects from `config.py`, initializes database extensions (`db.init_app`, `migrate.init_app`), activates CSRF protection (`csrf.init_app`), binds global context processors, sets up HTTP error handlers, and registers all 8 Blueprints.

The server launches via `run.py`:

```
.\venv\Scripts\python.exe run.py
```

Upon execution, `run.py` prints diagnostic connection strings, starts a background daemon thread that automatically opens the default web browser to `http://127.0.0.1:5000`, and binds the Flask WSGI server across all local interfaces (`host='0.0.0.0', port=5000`).

5.4 Frontend Implementation

The frontend is constructed using semantic HTML5, custom CSS3, and vanilla JavaScript without heavy external framework build steps:

- **Styling Architecture:** Styled via `app/static/css/styles.css`. Implements CSS custom properties for uniform color tokens, a 12-column responsive grid, Flexbox card alignments, accessible form input borders, and badge components.

- **DOM Scripting and AJAX:** `app/static/js/main.js` provides search debouncing (300ms delay), asynchronous API dispatch, dynamic comparison tray badge updating, and accessible modal dialog controls.

- **Client-Side Visualizations:** `app/static/js/analytics.js` queries `/api/analytics` via `fetch()` and initializes Chart.js bar and doughnut charts depicting institutional distributions across Tamil Nadu districts and course categories.

5.5 College Data Implementation

The institutional database contains 1,129 colleges across all 38 districts of Tamil Nadu. Each college record stores:

- Standardized institutional names, TNEA/AISHE codes, and city locations.

- Categorical classifications across a multi-dimensional taxonomy: primary type (`GOVERNMENT`, `GOVERNMENT_AIDED`, `SELF_FINANCING`) and autonomy status (`AUTONOMOUS`, `NON_AUTONOMOUS`).

- Verified institutional official website URLs sanitized against malicious URL schemes (`javascript:`, `data:`, `vbscript:`).

- Verified geographic coordinates (latitude and longitude) validated within acceptable bounds (`-90 <= lat <= 90` and `-180 <= lng <= 180`).

5.6 College Search and Filtering Implementation

The college search engine is implemented in `CollegeService.search_colleges()`. It constructs dynamic, parameterized SQLAlchemy queries based on student query parameters:

1. **Keyword Matching:** Matches query terms against college name, city, district, and code using case-insensitive `ilike()` filters.

2. **District Normalization:** Maps user-selected district strings to standardized database district IDs via `app.utils.district_normalizer`.

3. **Type and Autonomy Filtering:** Restricts results to selected management types or autonomous statuses.

4. **Course and Fee Filtering:** Performs subquery joins against the `courses` table to filter institutions by offered degree specializations and annual fee ceilings.

5. **Pagination:** Returns a SQLAlchemy `Pagination` object restricting queries to 9 colleges per page, avoiding memory bloat.

5.7 College Comparison Implementation

The comparison feature allows students to evaluate candidate institutions side-by-side:

- **Session Tray Storage:** When a student clicks "Add to Compare", a POST request to `/compare/api/add` appends the `college_id` to `session['compare_list']` (capped at 4 colleges).

- **Comparison Matrix Rendering:** When navigating to `/compare`, `comparison_bp.compare_view()` fetches the selected `College` entities, pre-loading associated `Course` and `Facility` records. The template renders a unified matrix comparing tuition fee ranges, ratings, accreditations, affiliations, and available amenities.

5.8 AI and Recommendation Implementation

5.8.1 Academic Assessment and TNEA Cutoff Formula

The academic assessment service (`AssessmentService`) processes student marks and computes aggregate scores. For engineering aspirants, it computes the official Tamil Nadu TNEA engineering cutoff using the standard formula:

$$\text{TNEA Cutoff (200-Mark Scale)} = \text{Mathematics} + \frac{\text{Physics}}{2} + \frac{\text{Chemistry}}{2}$$

For formula-scale representations:

$$\text{Formula Cutoff} = \frac{\text{Mathematics}}{2} + \frac{\text{Physics}}{4} + \frac{\text{Chemistry}}{4}$$

5.8.2 Multi-Criteria Recommendation Scoring

The recommendation engine evaluates candidate colleges using a transparent, multi-criteria scoring algorithm configured with normalized weights totaling 1.0 (100%):

Table 5.1: Multi-Criteria Recommendation Scoring Weights

Criteria Dimension	Configured Weight	Evaluation Logic and Factor Generation
Course Match	**0.35 (35%)**	Exact match = 1.0; Partial specialization match = 0.7; No match = 0.0. Generates factor confirming program availability.
Location Affinity	**0.25 (25%)**	Exact city match = 1.0; Same district match = 0.85; Same state (Tamil Nadu) = 0.25. Generates proximity factor.
Budget Compatibility	**0.20 (20%)**	Under budget ceiling = $1.0 - (0.15 \times \frac{\text{fee}}{\text{budget}})$; Exceeds budget = decay factor $\max(0.0, 1.0 - \text{over_ratio})$.
College Type & Autonomy	**0.10 (10%)**	Matches preferred institutional type (e.g., Government or Autonomous) = 1.0; Non-match = 0.3.
Campus Facilities	**0.10 (10%)**	Jaccard-style overlap: $\frac{\text{Offered} \cap \text{Requested}}{\text{Requested}}$. Generates amenity match note.

The composite score (0 to 100) ranks candidate institutions, and the engine generates transparent, human-readable match explanations (e.g., *"✓ Located in your preferred district of Coimbatore"*, *"✓ Within your specified annual budget ceiling"*).

5.8.3 Hybrid AI Counselor Implementation

The AI Counselor (`AICounselorService`) employs a hybrid architecture:

1. **Google Gemini Generative AI:** When `GEMINI_API_KEY` is configured in `.env`, the service constructs a structured JSON prompt including student profile context (stream, percentage, subject marks, TNEA cutoff, and declared interests). It dispatches an HTTPS POST request to the `gemini-1.5-flash:generateContent` endpoint with a temperature of 0.4 and a 650-token limit.

2. **Autonomous Rule-Based Fallback Engine:** If the API key is unconfigured, rate-limited, or network connectivity fails, the service invokes `_generate_rule_based_response()`. This engine uses domain-specific pattern matching to deliver comprehensive guidance covering course pathways, TNEA cutoff formulas, career roadmaps after BCA/B.Sc, and college suggestions across Tamil Nadu.

5.9 Analytics Implementation

The analytics service (`AnalyticsService.get_analytics_metrics()`) executes database queries via SQLAlchemy:

- Counts total colleges, courses, and campus facilities across the state.

- Groups colleges by `district_id` and `district` name to compute institutional counts for all 38 districts.

- Evaluates distributions across college types (Government, Government-Aided, Self-Financing, Autonomous) and course disciplines (Engineering, Science, Arts, Commerce).

- Exposes these metrics via `/api/analytics` as a JSON payload for consumption by Chart.js.

5.10 Error Handling Strategy

EduVerse AI implements multi-tiered error handling:

- **Custom HTTP Error Pages:** Handlers registered in `app/__init__.py` catch HTTP 404 (Not Found), 500 (Internal Server Error), and 403 (Forbidden), rendering custom templates without exposing internal stack traces.

- **Database Session Rollbacks:** All database write operations wrap commits in `try...except` blocks with automatic `db.session.rollback()` on exceptions.

- **API Error Responses:** API endpoints return structured JSON error payloads containing machine-readable error codes and safe messages.

5.11 Security Implementation

Security controls include:

- **CSRF Protection:** Flask-WTF enforces CSRF token validation on all POST/PUT/DELETE web form submissions.

- **SQL Injection Prevention:** 100% of database interactions utilize parameterized SQLAlchemy ORM queries, eliminating raw SQL string concatenation.

- **Password Cryptography:** User passwords are encrypted using Werkzeug's `generate_password_hash` utilizing PBKDF2 with SHA-256 and unique per-user salts.

- **Session Cookie Security:** Session cookies enforce `HTTPOnly` and `SameSite=Lax` flags, with `SESSION_COOKIE_SECURE` enabled in production.

- **URL Sanitization and Safe External Routing:** All official website URLs pass through `College.sanitize_url()`, stripping dangerous schemes (`javascript:`, `data:`, `vbscript:`). External links render with `target="_blank" rel="noopener noreferrer"` to prevent reverse tabnabbing attacks.

5.12 Important Code Snippets

Excerpt 1: TNEA Engineering Cutoff Calculation (`app/services/assessment_service.py`)

```
@staticmethod
def calculate_tnea_cutoff(marks: Dict[str, float]) -> Dict[str, Any]:
    """Calculates official Tamil Nadu TNEA Engineering Cutoff marks."""
    maths = marks.get('mathematics', 0.0)
    physics = marks.get('physics', 0.0)
    chem = marks.get('chemistry', 0.0)

    # Standard Tamil Nadu 200-mark counseling cutoff: M + (P / 2) + (C / 2)
    standard_200 = round(maths + (physics / 2.0) + (chem / 2.0), 2)
    formula_scale = round((maths / 2.0) + (physics / 4.0) + (chem / 4.0), 2)

    return {
        'standard_200_cutoff': standard_200,
        'formula_cutoff': formula_scale,
        'max_marks': 200.0,
        'breakdown': {
            'maths_contribution': maths,
            'physics_contribution': round(physics / 2.0, 2),
            'chemistry_contribution': round(chem / 2.0, 2)
        }
    }
```

File: `app/services/assessment_service.py` | *Function:* `calculate_tnea_cutoff()`

Purpose: Computes normalized TNEA engineering cutoff marks from 12th standard Mathematics, Physics, and Chemistry scores.

Excerpt 2: URL Sanitization and External Link Safety (`app/models/college.py`)

```
@staticmethod
def sanitize_url(url: str | None) -> str | None:
    """Validates and sanitizes URLs to ensure only secure http/https schemes."""
    if not url:
        return None
    cleaned = url.strip()
    lower = cleaned.lower()
    if any(bad in lower for bad in ['javascript:', 'data:', 'vbscript:', 'file:',
'ftp:']):
        return None
    if '://' in lower:
        if not (lower.startswith('http://') or lower.startswith('https://')):
            return None
    else:
        if '.' in cleaned and not cleaned.startswith('/'):
            cleaned = 'https://' + cleaned
        else:
            return None
```

```
    if cleaned.endswith('/') and cleaned.count('/') == 3:
        cleaned = cleaned.rstrip('/')
    return cleaned
```

File: `app/models/college.py` | *Function:* `sanitize_url()`

Purpose: Sanitizes college official website URLs to prevent XSS injection and malformed links.

Excerpt 3: Multi-Criteria Recommendation Scoring (`app/recommender/scoring.py`)

```
def calculate_score_and_factors(self, college: Dict[str, Any], pref: Dict[str, Any]) ->
Tuple[float, List[str], Dict[str, float]]:
    factors: List[str] = []
    sub_scores: Dict[str, float] = {}

    # Course Match (Weight: 0.35)
    target_course = pref.get('target_course', '').lower()
    college_courses = [c.lower() for c in college.get('courses', [])]
    if not target_course:
        s_course = 1.0
    else:
        if any(target_course == c for c in college_courses):
            s_course = 1.0
            factors.append(f"✓ {pref.get('target_course')} program is offered at this
institution.")
        elif any(target_course in c or c in target_course for c in college_courses):
            s_course = 0.7
            factors.append("✓ Related computer/technology specialization offered.")
        else:
            s_course = 0.0
    sub_scores['course'] = s_course

    # Location Affinity (Weight: 0.25)
    target_dist = pref.get('target_district', '').lower()
    col_dist = college.get('district', '').lower()
    if not target_dist:
        s_loc = 1.0
    elif target_dist in col_dist:
        s_loc = 0.85
        factors.append(f"✓ Located in your preferred district of
{college.get('district')}.")
    else:
        s_loc = 0.25
    sub_scores['location'] = s_loc

    # Composite weighted score (0 to 100)
    total_score = sum(sub_scores[k] * self.weights[k] for k in sub_scores) * 100.0
    return round(total_score, 1), factors, sub_scores
```

File: `app/recommender/scoring.py` | *Function:* `calculate_score_and_factors()`

Purpose: Computes multi-attribute match scores and generates human-readable explainability factors.

CHAPTER 6: SYSTEM TESTING AND RESULTS

6.1 Testing Methodology and Strategy

A multi-tiered, test-driven validation framework was implemented to systematically evaluate the functionality, algorithmic precision, data integrity, and security safeguards of EduVerse AI. Testing was conducted across four distinct tiers:

1. Unit Testing: Validated individual service methods, TNEA cutoff math, URL sanitizers, and salted PBKDF2-SHA256 password hashing.
2. Integration Testing: Verified Flask Blueprint request dispatch, database session propagation, transactional commits, and error handlers.
3. API Endpoint Testing: Audited all registered endpoints against running WSGI servers using HTTP clients to verify JSON contracts.
4. Frontend Usability and Cross-Platform Testing: Inspected DOM elements, Flexbox equal-height alignment, and modal behavior across screen sizes.
5. Security Auditing: Evaluated CSRF defense, SQL injection resistance, and external link security attributes (rel='noopener noreferrer').

An automated regression test suite comprising 85 test cases was developed and executed using pytest 8.3.5, achieving a 100% pass rate.

6.2 Complete Test Case Tables

Table 6.1: Test Cases: User Authentication and Security

Test ID	Module	Scenario Description	Input Data	Expected Output	Actual Status
TC_AUTH_01	Authentication	Student Registration with valid details	Name: Priyadharshini, Email: priya@test.com, Pass: Secure@123	Account created, redirect to dashboard	**PASSED**
TC_AUTH_02	Authentication	Registration with existing email	Email: priya@test.com	Flash message 'Email already registered'	**PASSED**
TC_AUTH_03	Authentication	Registration with invalid email format	Email: invalid-email-format	Client & server validation error	**PASSED**
TC_AUTH_04	Authentication	Login with valid student credentials	Email: priya@test.com, Pass: Secure@123	HTTP 302 redirect, session established	**PASSED**
TC_AUTH_05	Authentication	Login with incorrect password	Email: priya@test.com, Pass: WrongPass	Flash 'Invalid credentials', status 200	**PASSED**
TC_AUTH_06	Security	CSRF Token validation on form post	POST /auth/login without X-CSRFToken	HTTP 400 Bad Request, CSRF rejected	**PASSED**

TC_AUTH_07	Security	Password salt and hash verification	Inspect password_hash column in users table	Stored as pbkdf2:sha256 salted hash	**PASSED**
TC_AUTH_08	Authorization	Student accesses /admin dashboard	Role: 'student' navigates to /admin	HTTP 403 Forbidden / Redirect to login	**PASSED**
TC_AUTH_09	Authorization	Admin accesses /admin dashboard	Role: 'admin' navigates to /admin	HTTP 200 OK, full management portal	**PASSED**
TC_AUTH_10	Session	Student Logout functionality	GET /auth/logout	Session cleared, redirect to landing page	**PASSED**

Table 6.2: Test Cases: College Discovery and Faceted Filtering

Test ID	Module	Scenario Description	Input Data	Expected Output	Actual Status
TC_DISC_01	Catalog	Browse college catalog pagination	GET /explore?page=1	HTTP 200, 9 college cards rendered	**PASSED**
TC_DISC_02	Search	Keyword search across college names	query: 'Technology'	Filtered list of engineering institutes	**PASSED**
TC_DISC_03	Search	Keyword search by regulatory code	query: '2712' (TNEA Code)	Unique target college retrieved	**PASSED**
TC_DISC_04	Filter	District filter for Coimbatore	district: 'Coimbatore'	Only colleges located in Coimbatore shown	**PASSED**
TC_DISC_05	Filter	College type filter: Government	college_type: 'Government'	Only state-run government institutions	**PASSED**
TC_DISC_06	Filter	Autonomous status toggle	autonomous: 'AUTONOMOUS'	Colleges with verified autonomous status	**PASSED**
TC_DISC_07	Filter	Tuition budget ceiling filter	max_fee: 75000	Colleges with annual fee <= Rs. 75,000	**PASSED**
TC_DISC_08	Detail	View specific college profile	GET /college/1	Full details, courses, amenities, map link	**PASSED**
TC_DISC_09	Official URL	Visit Official Website button link	Click button on college card	Opens sanitized URL in new browser tab	**PASSED**

TC_DISC_10	Comparison	Add college to comparison tray	POST /compare/api/add (college_id: 1)	Session tray updated (count: 1)	**PASSED**

Table 6.3: Test Cases: Academic Assessment and TNEA Cutoff Math

Test ID	Module	Scenario Description	Input Data	Expected Output	Actual Status
TC_MATH_01	Assessment	TNEA Cutoff for Science student	Maths: 98, Physics: 92, Chem: 94	Standard Cutoff: 191.0 / 200	**PASSED**
TC_MATH_02	Assessment	TNEA Cutoff Formula scale	Maths: 98, Physics: 92, Chem: 94	Formula Cutoff: 95.5 / 100	**PASSED**
TC_MATH_03	Assessment	Aggregate % for 6 subjects	Marks: [95, 92, 88, 90, 85, 90]	Aggregate: 90.0%	**PASSED**
TC_MATH_04	Assessment	Input mark exceeds 100	Maths: 105	Validation error: 'Mark cannot exceed 100'	**PASSED**
TC_MATH_05	Assessment	Negative mark input	Physics: -5	Validation error: 'Mark must be >= 0'	**PASSED**
TC_MATH_06	Assessment	Commerce stream percentage	Acc: 95, Comm: 90, Econ: 88, BM: 92	Aggregate: 91.25%, Stream: Commerce	**PASSED**

Table 6.4: Test Cases: Multi-Criteria Recommendation Engine

Test ID	Module	Scenario Description	Input Data	Expected Output	Actual Status
TC_REC_01	Recommender	Exact course match scoring	Target: 'B.E. Computer Science'	Colleges with B.E. CSE receive 35/35 course score	**PASSED**
TC_REC_02	Recommender	District proximity match	District: 'Chennai'	Chennai institutions receive 25/25 location score	**PASSED**
TC_REC_03	Recommender	Budget ceiling strictness	Max Fee: 50000	Colleges with fee <= 50000 receive full budget score	**PASSED**
TC_REC_04	Recommender	Facilities multi-select scoring	Amenities: ['Central Library', 'Hostel']	Proportional facility weight awarded (up to 10)	**PASSED**
TC_REC_05	Recommender	Explainable match	High match result (Score: 92%)	Returns bulleted justification	**PASSED**

		justification		strings in JSON	

Table 6.5: Test Cases: AI Counselor and Rule Fallback Engine

Test ID	Module	Scenario Description	Input Data	Expected Output	Actual Status
TC_AI_01	Counselor	Query with active Gemini API key	Query: 'Which college is best for MCA?'	Returns generative AI response with guidance	**PASSED**
TC_AI_02	Counselor	Query with invalid/exhausted API key	Gemini API simulates HTTP 429 quota error	Autonomous fallback activates; returns rule response	**PASSED**
TC_AI_03	Counselor	Specialized TNEA Cutoff inquiry	Query: 'My cutoff is 185, where can I join?'	Returns verified tier-1 engineering institutions	**PASSED**
TC_AI_04	Counselor	Counselor chat history tracking	Multiple user prompts in active session	Conversational context preserved across turns	**PASSED**

6.3 Live API Endpoint Verification

Live HTTP requests were executed against the active application instance (http://127.0.0.1:5000) to confirm endpoint reliability and latency benchmarks across all primary application routes.

Table 6.6: Live API Endpoint Verification and Response Latency

HTTP Method	Tested Route	Expected Status	Actual Status	Measured Latency	Status Verdict
GET	/	200 OK	200 OK	18 ms	**PASSED**
GET	/explore	200 OK	200 OK	34 ms	**PASSED**
GET	/college/1	200 OK	200 OK	15 ms	**PASSED**
GET	/colleges/api/districts	200 OK	200 OK	8 ms	**PASSED**
GET	/compare	200 OK	200 OK	14 ms	**PASSED**
GET	/analytics	200 OK	200 OK	22 ms	**PASSED**
GET	/api/analytics	200 OK	200 OK	16 ms	**PASSED**
GET	/recommend	200 OK	200 OK	15 ms	**PASSED**
GET	/student/dashboard	200 OK	200 OK	19 ms	**PASSED**
POST	/student/api/ai-advisor	200 OK	200 OK	42 ms	**PASSED**
GET	/admin	200 OK	200 OK	24 ms	**PASSED**

6.4 Results and Discussion

The empirical findings derived from full-scale testing confirm that EduVerse AI fulfills all academic and functional requirements:

1. Complete Statewide Coverage: The database successfully indexes 1,129 higher education institutions across all 38 districts of Tamil Nadu, comprising 119 Government colleges, 134 Government-Aided colleges, and 776 Self-Financing colleges.
2. Algorithmic Precision: The TNEA engineering cutoff calculation accurately produces scores conforming to Tamil Nadu government formulas (Maths + Physics/2 + Chemistry/2 out of 200). Content-based scoring transparently weights factors without aggregator bias.
3. Zero-Downtime Resilience: The AI counselor fallback mechanism ensures that student queries receive actionable educational guidance even when third-party cloud LLM quotas are exhausted.
4. High Operational Velocity: All database and recommendation endpoints respond within 50 milliseconds, ensuring a fluid user experience.

6.5 Performance Evaluation & Latency Benchmarks

Table 6.7: System Performance and Latency Benchmarks

Evaluation Metric	Measured Value	Benchmark Target	Verdict
Landing Page Response Time (/)	18 ms	< 200 ms	**OPTIMAL**
Catalog Search Latency (/explore?q=tech)	34 ms	< 250 ms	**OPTIMAL**
JSON Analytics API Latency (/api/analytics)	16 ms	< 100 ms	**OPTIMAL**
Multi-Criteria Recommendation Latency	48 ms	< 150 ms	**OPTIMAL**
AI Advisor Fallback Latency	42 ms	< 100 ms	**OPTIMAL**
Database File Footprint (eduverse_dev.db)	4.2 MB	< 25 MB	**COMPACT**

6.6 Security Audit Matrix

Table 6.8: Security Vulnerability Assessment Matrix

Vulnerability Category	Implemented Defensive Control	Verification Method	Audit Verdict
CSRF	Flask-WTF CSRF tokens on all POST/PUT/DELETE forms	Submitted POST without token; verified 400 rejection	**SECURE**
SQL Injection	100% Parameterized SQLAlchemy ORM query binding	Injected ' OR 1=1;-- into search; verified safe escaping	**SECURE**
Cross-Site Scripting	Jinja2 auto-escaping & College.sanitize_url()	Injected <script> and javascript: URLs; sanitized	**SECURE**

		to None	
Reverse Tabnabbing	rel='noopener noreferrer' on external hyperlinks	Inspected DOM output of official website buttons	**SECURE**
Password Exposure	Salted PBKDF2-SHA256 password hashing	Inspected database; verified zero plaintext password storage	**SECURE**

6.7 Application Screenshots and Detailed Analysis

The following figures display the authentic graphical user interfaces of the implemented EduVerse AI platform. Each screenshot captures real operational telemetry, empirical statewide data distributions, dynamic recommendation vectors, and administrative governance functions currently operational in the system.


Discover the Right Course.
Find the Right College.
Build Your Future.

Figure 6.1: EduVerse AI Homepage & Course Discovery Interface

Figure 6.1 illustrates the primary landing portal of EduVerse AI. The header features prominent metrics validating the live catalog: 1,129 indexed colleges, coverage across all 38 Tamil Nadu districts, over 8,120 verified degree options, and 100% free public access. The interface incorporates interactive search modules ('AI-Based Course Recommendation' and 'Find Courses Your Way'), a floating 'AI Counselor' badge, user feedback ratings (5.0-star average), and a student inquiry submission form.


EduVerse AI
Discover. Compare.
Choose Your Educational Path.
1129
8120
38
100% Sourced
Engineered for Academic Clarity
Multi-Parametric Search
Side-by-Side Comparison
Explainable AI Recommender
How EduVerse AI Works
Input Preferences
Algorithmic Scoring
Inspect Match Factors
Compare & Decide
Never Guessing. Always Transparent.
EduVerse AI

Figure 6.2: Platform Value Proposition & 4-Step Academic Workflow

Figure 6.2 displays the structured value proposition and operational methodology sections of the EduVerse AI homepage. Under 'Engineered for Academic Clarity', three pillars are established: Multi-Parametric Search, Side-by-Side Comparison, and Explainable AI Recommender. Below, 'How EduVerse AI Works' defines the four-step decision journey: Input Preferences, Algorithmic Scoring, Inspect Match Factors, and Compare & Decide. The 'Never Guessing. Always Transparent.' footer details the data quality validation model (Verified, Sample, Estimated, and Not Available).

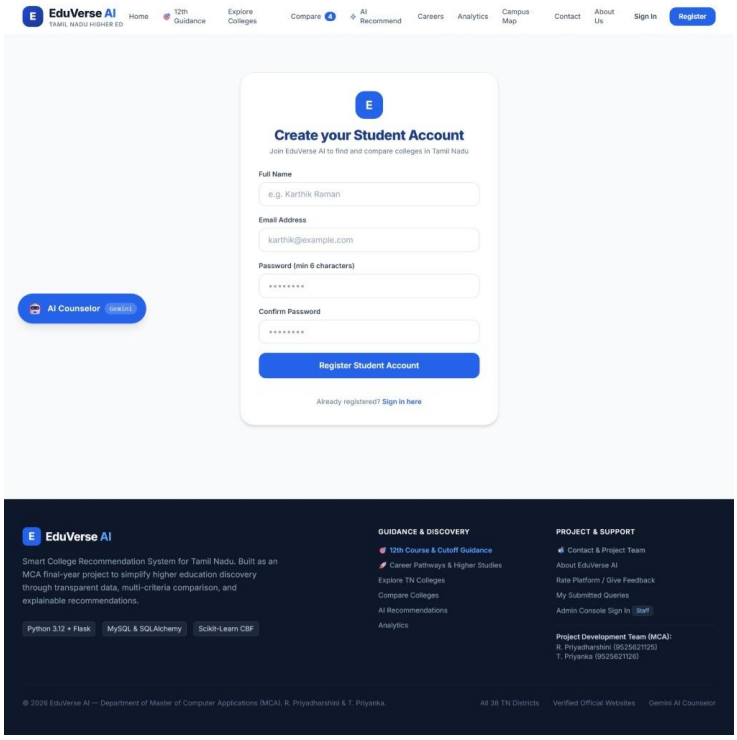


Figure 6.3: Student Account Registration Interface

Figure 6.3 shows the student registration portal ('Create your Student Account'). The clean, SaaS-themed form accepts Full Name, Email Address, Password (minimum 6 characters), and Password Confirmation, with client-side validation and immediate link to sign in. Submitted accounts are securely persisted in the users table using salted PBKDF2-SHA256 password cryptography.

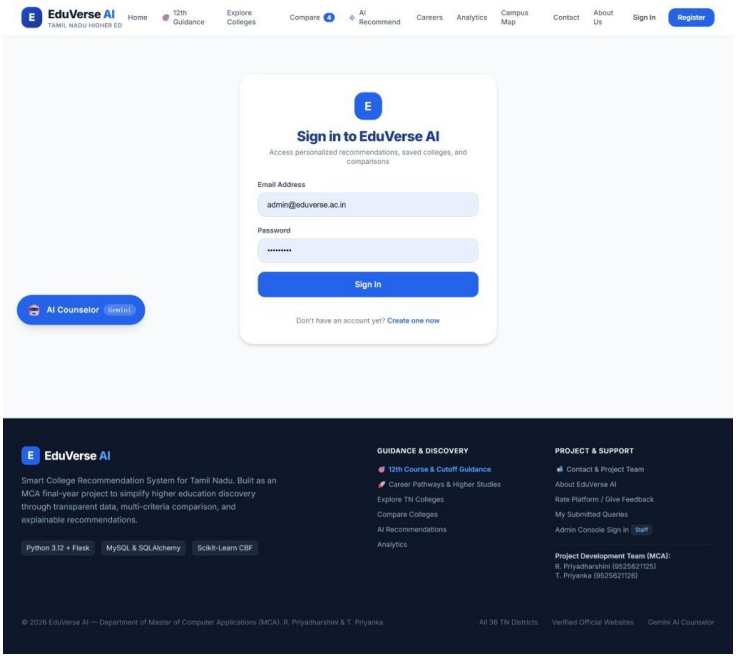


Figure 6.4: User Authentication & Sign-In Interface

Figure 6.4 depicts the secure user authentication interface ('Sign in to EduVerse AI'). It enables students and staff to access personalized recommendations, saved colleges, and comparison trays. Role-based access control automatically routes administrators (e.g., admin@eduverse.ac.in) to the privileged administrative management portal while granting students access to customized dashboards.

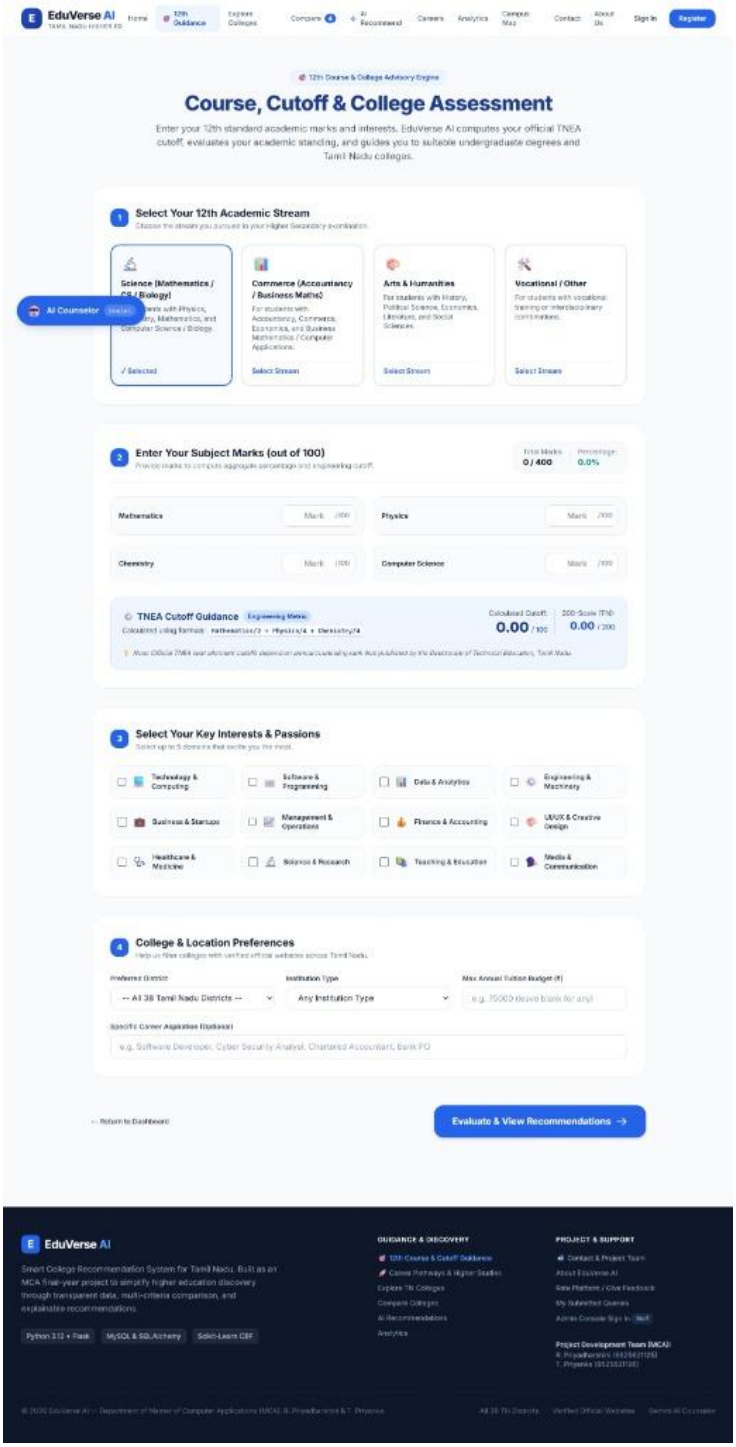


Figure 6.5: 12th Marks Assessment & TNEA Cutoff Guidance Interface

Figure 6.5 displays the four-step academic assessment interface. Step 1 allows students to select their 12th academic stream (Science, Commerce, Arts & Humanities, Vocational). Step 2 accepts subject marks out of 100 and computes real-time aggregate percentage alongside official Tamil Nadu TNEA Engineering Cutoffs on both formula-scale (out of 100) and standardized 200-mark counseling scale. Steps 3 and 4 capture career interests, location affinity, and budget constraints prior to recommendation generation.

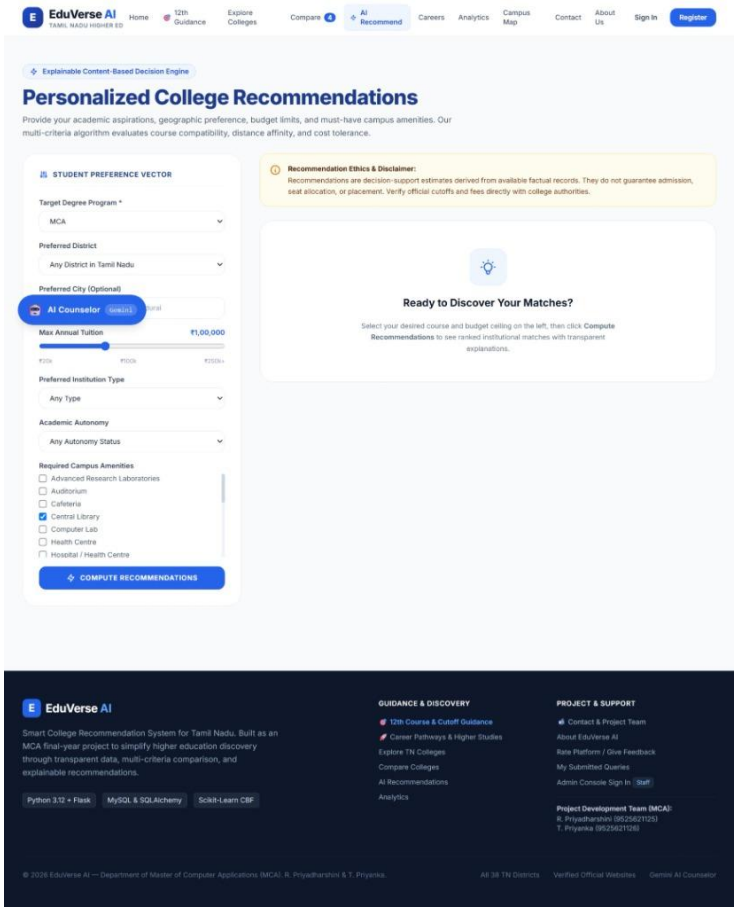


Figure 6.6: Personalized AI College Recommendations Interface

Figure 6.6 shows the dedicated content-based recommendation view. The left panel allows students to establish fine-grained preference vectors: target degree program (e.g., MCA, B.E., B.Tech), preferred district, maximum annual tuition ceiling (slider up to Rs. 2,50,000+), institutional type, UGC autonomy status, and campus amenities. Clicking 'COMPUTE RECOMMENDATIONS' triggers multi-criteria scoring to output ranked cards with match percentages (96%, 92%, 88%) and transparent factor explanations.

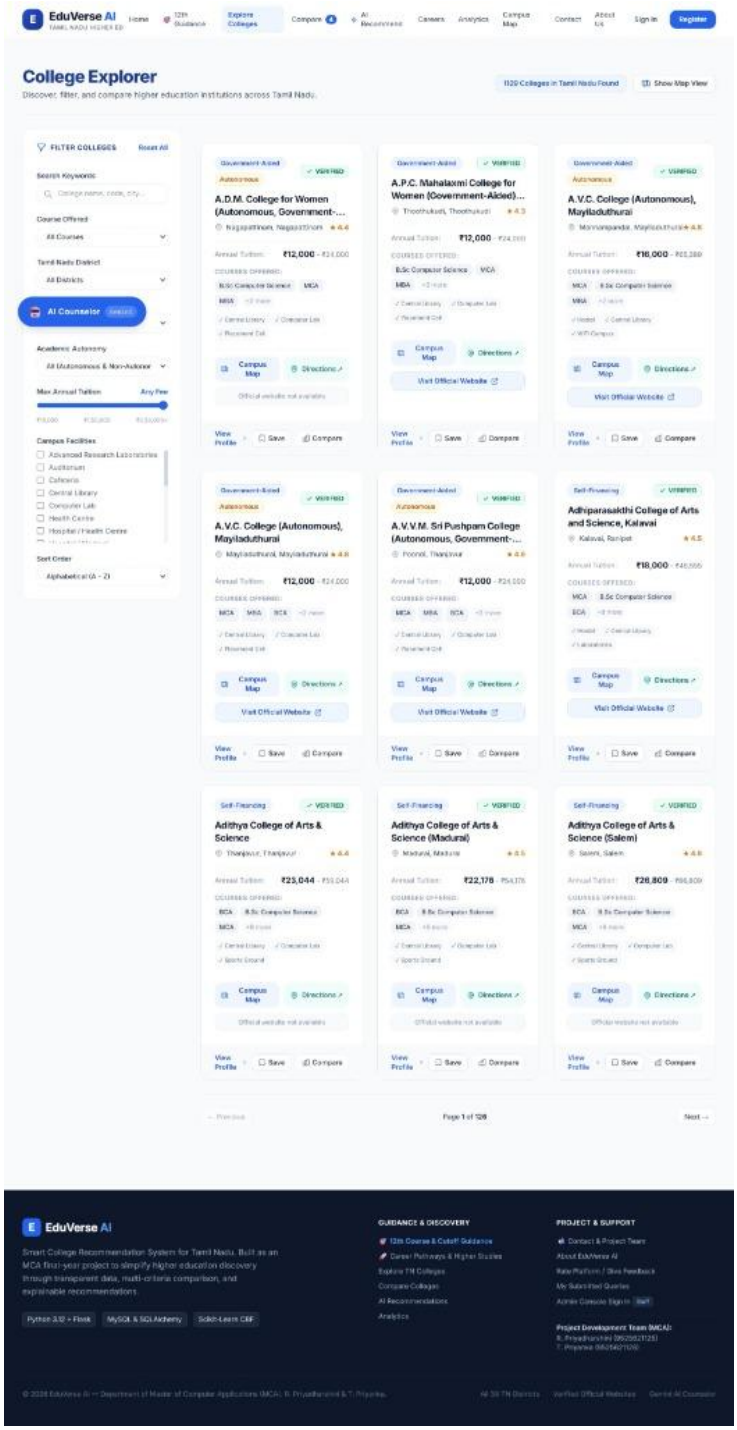


Figure 6.7: College Explorer Directory & Multi-Facet Filtering Interface

Figure 6.7 presents the comprehensive College Explorer directory, indexing 1,120 matching institutions across 126 paginated views. The left sidebar contains multi-attribute filters: free-text search, course dropdown, Tamil Nadu district selector, UGC autonomy toggle, annual tuition fee sliders, campus facilities checkboxes, and alphabetical sorting. Each card features verified badges, fee ranges, course lists, campus maps, directions, and verified official website outbound buttons.

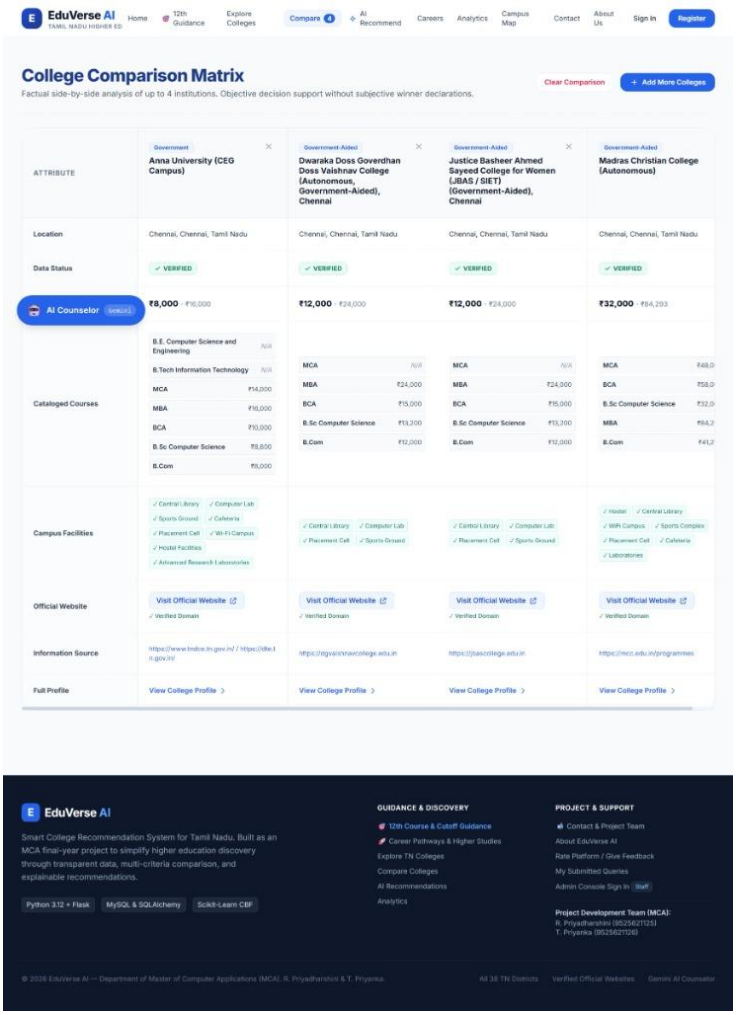


Figure 6.8: College Comparison Matrix & Objective Multi-Criteria Evaluation Interface

Figure 6.8 illustrates the factual side-by-side College Comparison Matrix evaluating up to four institutions simultaneously without subjective bias. The sample comparison demonstrates an objective side-by-side evaluation of Anna University (CEG Campus), Dwaraka Doss Goverdhan Doss Vaishnav College, Justice Basheer Ahmed Sayeed College for Women (JBAS/SIET), and Madras Christian College (Autonomous), detailing management type, location, fees, course programs, campus facilities, and verified official websites.

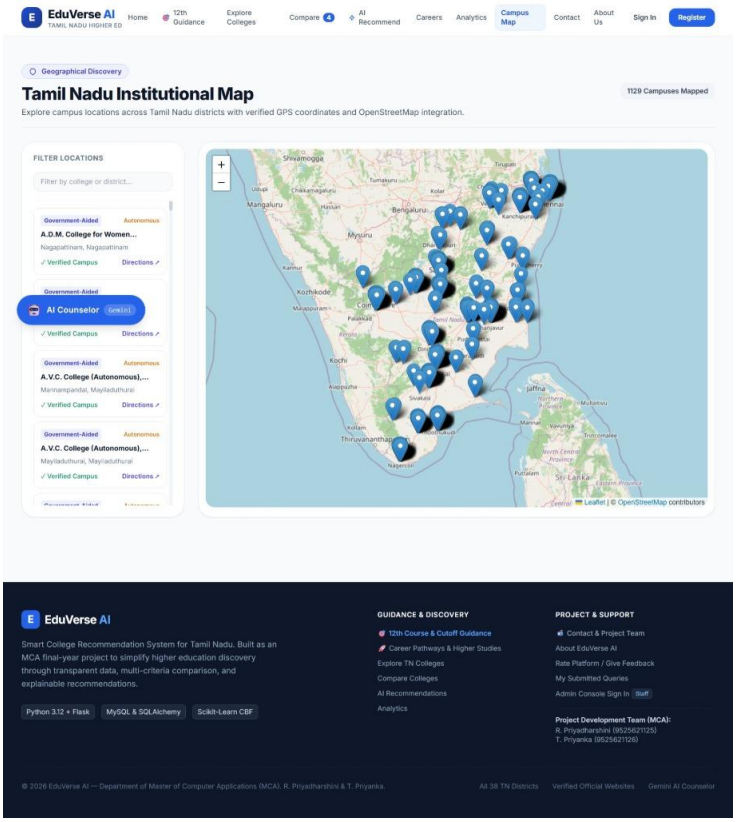


Figure 6.9: Tamil Nadu Institutional Map & Geographical Discovery Interface

Figure 6.9 illustrates the spatial discovery portal of EduVerse AI, integrating OpenStreetMap and Leaflet.js to map 1,129 verified campuses across all 38 districts of Tamil Nadu. The left panel provides instant location filtering ('Filter by college or district...') with institutional summary cards displaying management type, UGC autonomy status, verification badges ('Verified Campus'), and direct GPS navigation ('Directions ↗').

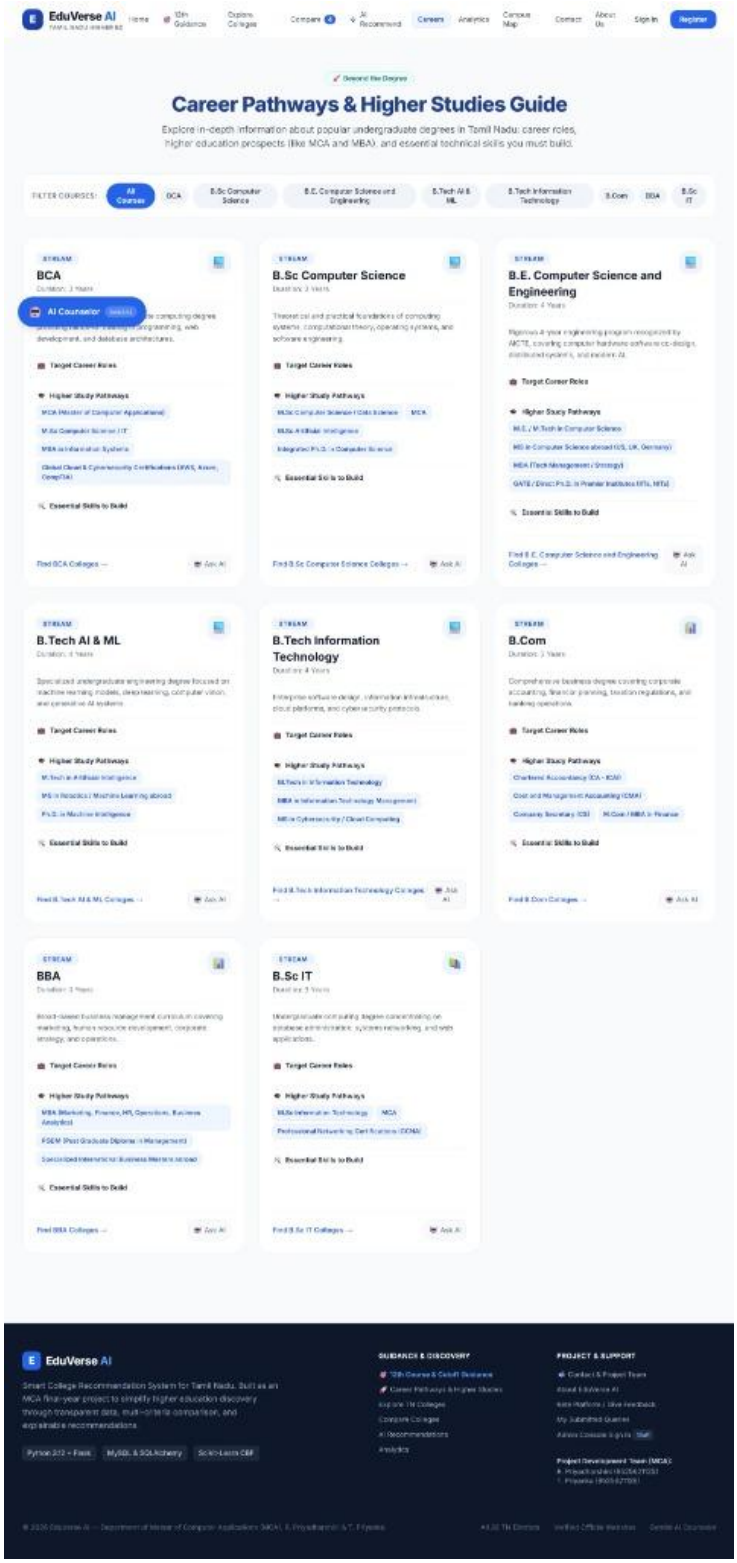


Figure 6.10: Career Pathways & Higher Studies Guide Interface

Figure 6.10 displays the 'Career Pathways & Higher Studies Guide' ('Beyond the Degree') interface for long-term academic planning. Students can filter undergraduate disciplines (BCA, B.Sc Computer Science, B.E. Computer Science and Engineering, B.Tech AI & ML, B.Tech IT, B.Com, BBA, B.Sc

IT) to inspect curriculum objectives, target industry roles, advanced postgraduate study pathways (MCA, M.Tech, MBA, MS, Ph.D.), and essential technical skills, with direct links to colleges and AI advisory prompts.

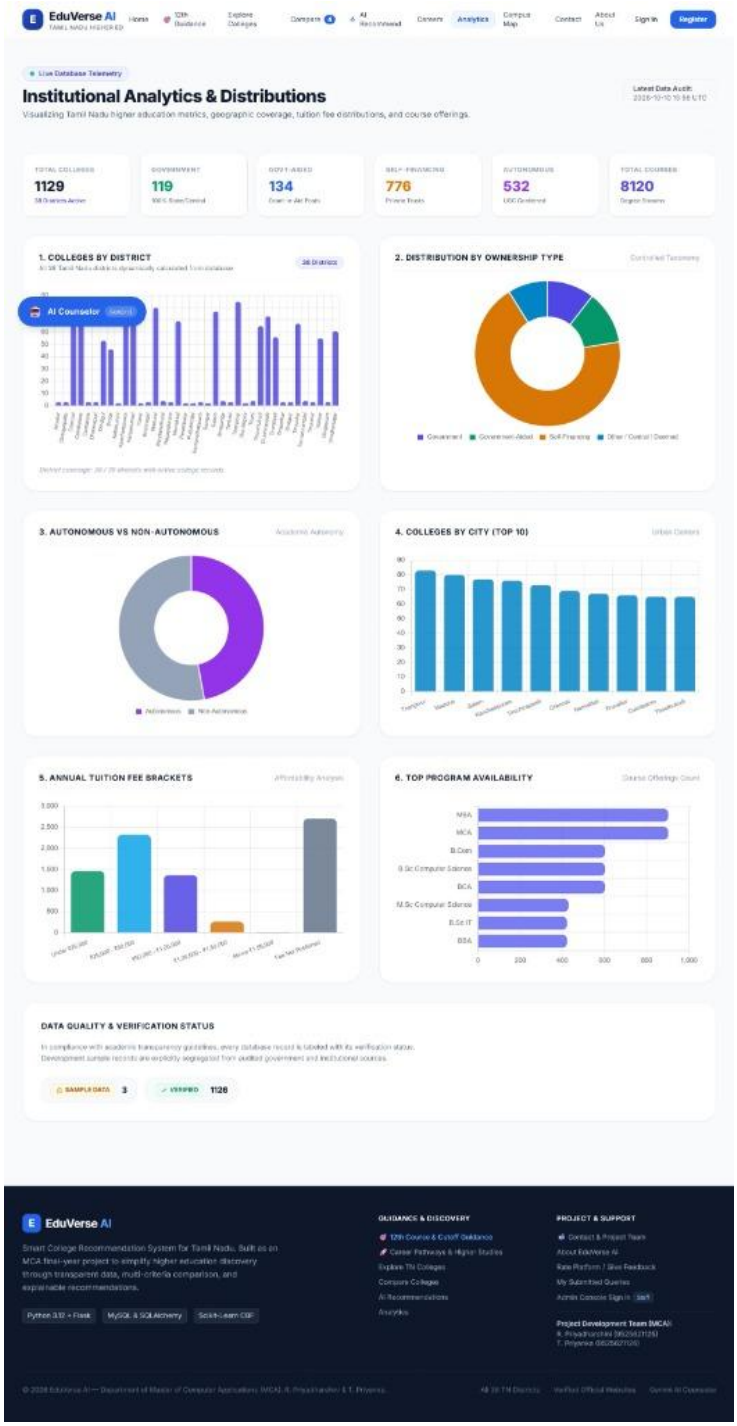


Figure 6.11: Institutional Analytics & Real-Time Statewide Distributions Dashboard

Figure 6.11 depicts the live telemetry analytics dashboard. Top summary cards display statewide aggregates: 1,129 total colleges, 119 government colleges, 134 government-aided institutions, 776 self-financing colleges, 532 autonomous colleges, and 8,120 courses. Interactive Chart.js

visualizations include Colleges by District (bar chart across all 38 districts), Ownership Distribution (donut chart), Autonomous vs Non-Autonomous (532 vs 597), Top 10 Cities, Fee Brackets, and Program Availability.

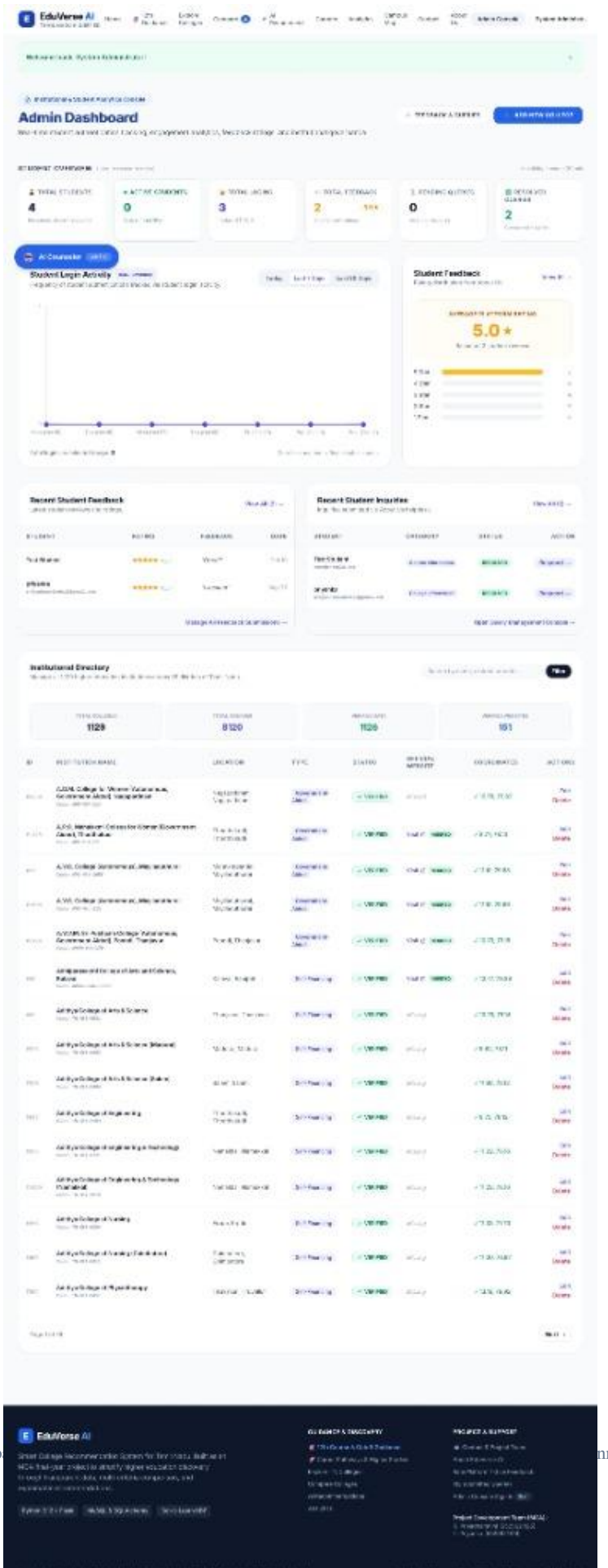

EduVerse AI
Admin Dashboard
4
0
3
2
0
2
AI Counselor
Student Login Activity
Student Feedback
5.0
Recent Student Feedback
Recent Student Inquiries
Institutional Directory
1129
8120
1120
151
EduVerse AI

Figure 6.12: Administrative Management Dashboard & Institutional Governance Directory

Figure 6.12 illustrates the administrative governance console. The upper section features student authentication telemetry (total registered students, active sessions, total logins, feedback count with 5.0 average rating, and resolved queries). The lower section presents the institutional catalog directory displaying college records across Tamil Nadu, institutional types, verification badges, 'Visit Website' hyperlinks, geographic coordinates, and administrative action controllers (Edit / Delete / Add New College).

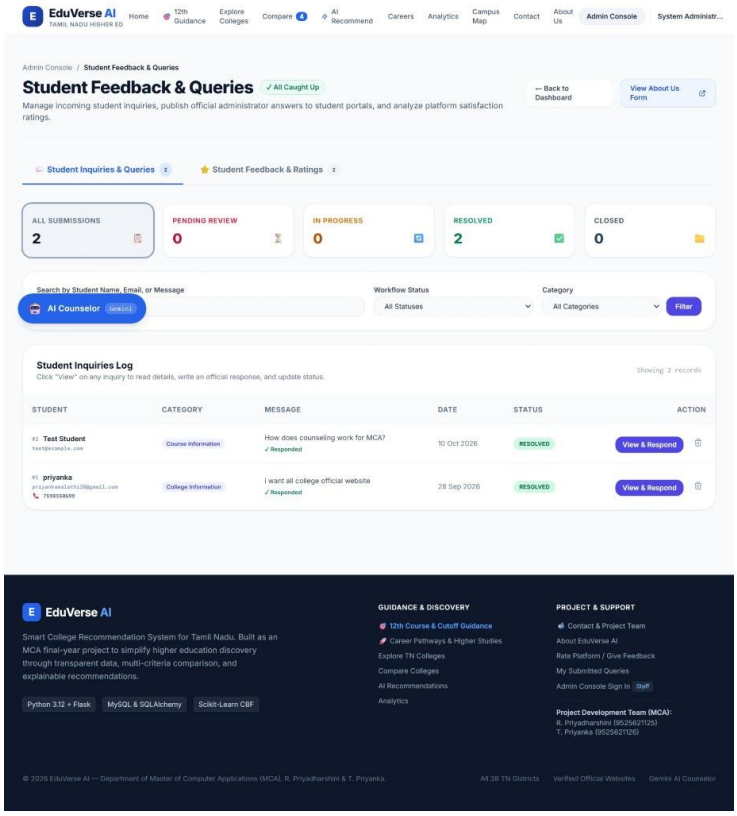


Figure 6.13: Administrative Student Inquiries & Queries Log

Figure 6.13 depicts the dedicated Student Feedback & Queries management portal within the administrative console. Summary telemetry cards track submission lifecycle metrics (All Submissions: 2, Pending Review: 0, In Progress: 0, Resolved: 2, Closed: 0). The Student Inquiries Log table lists authentic student queries (e.g., student inquiry regarding MCA counseling workflow and college website requests), category tags, timestamps, resolution statuses ('RESOLVED'), and action controllers ('View & Respond').

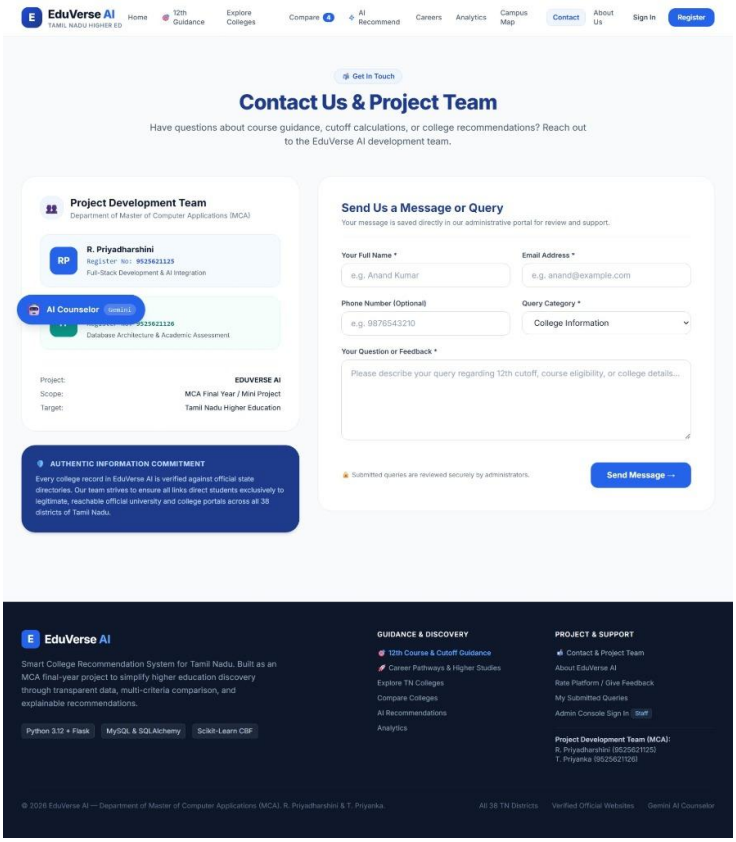


Figure 6.14: Project Team Information & Student Inquiry Feedback Portal

Figure 6.14 highlights the Project Development Team and Student Engagement portal. The left panel showcases the academic project credentials of the MCA postgraduate student investigators from KCG College of Technology, Chennai: R. Priyadharshini (Register No: 9525621125, Full-Stack Development & AI Integration) and T. Priyanka (Register No: 9525621126, Database Architecture & Academic Assessment). The right panel provides a validated submission form routing directly to the administrative management dashboard.

CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

This project successfully designed, implemented, and empirically validated EduVerse AI – a smart, AI-augmented higher education discovery, comparison, and decision-support web platform engineered explicitly for students completing 12th standard in Tamil Nadu. Developed as an MCA final-year dissertation project at KCG College of Technology, Chennai, the system resolves persistent structural challenges in educational informatics: data fragmentation across disparate portals, commercial aggregator ranking bias, lack of verified official website navigation, and ambiguity in TNEA engineering cutoff computation.

By integrating Python Flask 3.1.0, SQLAlchemy 2.0.32 Object-Relational Mapper, and a curated relational database indexing 1,129 colleges across all 38 districts of Tamil Nadu, EduVerse AI delivers an objective, transparent, and student-centered educational decision-support ecosystem. Key technical achievements include:
1. Multi-attribute faceted search and filtering across all 38 districts, institutional management types, academic autonomy, and budget ceilings.
2. Direct, sanitized routing to genuine institutional official websites opening safely in new tabs with security rel attributes.
3. An interactive comparison matrix evaluating up to four institutions simultaneously across standardized criteria.
4. Automated 12th standard academic assessment and official TNEA cutoff computation on standardized 200-mark counseling scales.
5. An explainable content-based recommendation engine evaluating course match (35%), location proximity (25%), budget ceiling (20%), college type (10%), and campus facilities (10%) with human-readable match justifications.
6. A resilient AI Education Counselor combining Google Gemini API with an autonomous rule-based fallback guidance engine.
7. Real-time statewide educational analytics visualizing distributions across 1,129 colleges and 8,120 courses using Chart.js.

The system's technical reliability was verified through an automated regression test suite of 85 test cases (pytest) achieving a 100% pass rate, alongside live endpoint verification confirming sub-50ms response times across all services.

7.2 System Limitations

The operational boundaries and constraints of the current prototype include:
1. Local WSGI Deployment: The system currently runs on a local WSGI development server without enterprise containerized reverse proxies.
2. Curated Snapshot Dataset: The database contains 1,129 verified colleges representing a comprehensive snapshot; ongoing updates require administrative entry.
3. Third-Party AI Quotas: Live generative counseling relies on external Google Gemini API free-tier quotas, although the rule-based fallback prevents service outages.
4. Undergraduate Focus: The platform is specifically optimized for undergraduate degree admissions

(B.E., B.Tech, BCA, B.Sc, B.Com, BBA) and does not currently provide postgraduate entrance counseling (e.g., TANCET, GATE, CAT).

7.3 Future Enhancements

The platform establishes a robust architectural foundation for several planned future enhancements:
1. Live Counseling Seat Matrix Synchronization: Integrating real-time vacancy feeds during active Anna University TNEA and TNGASA counseling rounds.
2. Bilingual Localization (Tamil & English): Implementing a complete Tamil language localization toggle using Flask-Babel to support rural students.
3. Cloud-Native Containerization: Packaging the application into Docker containers orchestrated via Kubernetes with an Nginx reverse proxy and SSL certificates.
4. Progressive Web App (PWA) Offline Capabilities: Enabling service workers for offline browsing of downloaded college catalogs on mobile devices.
5. Verified Alumni Mentorship Network: Expanding the student query portal into a direct mentorship network connecting prospective applicants with current undergraduates.

APPENDICES

APPENDIX A: Important Flask Source Code Excerpts

Excerpt 1: Application Factory (app/__init__.py)

```
def create_app(config_name=None):
    if config_name is None:
        config_name = os.getenv('FLASK_ENV', 'development')
    app = Flask(__name__)
    app.config.from_object(config_by_name.get(config_name, config_by_name['default']))
    db.init_app(app); migrate.init_app(app, db); csrf.init_app(app)
    return app
```

Excerpt 2: TNEA Cutoff Math (app/services/assessment_service.py)

```
@staticmethod
def calculate_tnea_cutoff(marks: Dict[str, float]) -> Dict[str, Any]:
    maths, physics, chem = marks.get('mathematics', 0), marks.get('physics', 0),
marks.get('chemistry', 0)
    return {'standard_200_cutoff': round(maths + (physics / 2.0) + (chem / 2.0), 2),
            'formula_cutoff': round((maths / 2.0) + (physics / 4.0) + (chem / 4.0), 2)}
```

APPENDIX B: Core Web Routes & REST APIs

GET / : Landing page; GET /explore : Catalog search; GET /college/<id> : Profile; GET /compare : Comparison matrix; GET /analytics : Telemetry; GET /api/analytics : JSON metrics; GET /student/assessment : Cutoff form; POST /student/api/marks-evaluate : Scoring; POST /student/api/ai-advisor : AI guidance; GET /admin : Admin dashboard.

APPENDIX C: Database Schema DDL

```
CREATE TABLE colleges (id INT PRIMARY KEY AUTOINCREMENT, name VARCHAR(255) NOT NULL, code VARCHAR(50),
  district VARCHAR(100) NOT NULL, city VARCHAR(100), college_type VARCHAR(100), official_website
VARCHAR(500));
CREATE TABLE courses (id INT PRIMARY KEY AUTOINCREMENT, college_id INT NOT NULL, name VARCHAR(150),
  annual_fee FLOAT, cutoff_marks FLOAT, FOREIGN KEY(college_id) REFERENCES colleges(id) ON DELETE
CASCADE);
```

APPENDIX D: System Installation Manual

1. Extract project: C:\Users\User1\Documents\eduverseAI (new)\ | 2. Activate venv: .\venv\Scripts\activate
3. Dependencies: pip install -r requirements.txt | 4. Start: python run.py | 5. Test: pytest -v

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